

Romance never dies

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You only live once. Do what you love.

Show my great respect to my grandmother (Xiangfen Cai) and my parents.

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Chapter 1

Federated Learning

1.1 Theory

Some theoretical analysis about federated learning

- Intellectual property rights: A critical history (06 [867])
- Federated Optimization: Distributed Machine Learning (arXiv'16 [588])
- The Eu General Data Protection Regulation (gdpr) (2017 [1209])
- Toward fairness in data sharing (NEJM'16 [918])
- The california consumer privacy act (2018 [943])
- Convergence rate of FedAvg on NonIID is $\mathcal{O}(\frac{1}{T})$ (arXiv'19 [682])
- DNNs with batch norm are vulnerable to skewed label (ICML'20 [435])
- Lower bounds and optimal algorithms for personalized FL (NeurIPS'20 [390])
- Linearly converge in NonIID with part clients $\mathcal{O}(\frac{1}{\sqrt{nKT}} + \frac{1}{T})$ (ICLR'21 [1484])
- Linear speedup for FedAvg in statistics and system heter (ICLR'21 [994])
- Quantify and estimate influence over the model parameters (AAAI'21 [1464])
- Sharp lower bounds for homo and heter FedAvg (AISTAS'22 [325])
- Federated Minimax Optimization: improved convergence (ICML'22 [1055])
- Initial learning phase plays a critical role in FL (AAAI'22 [1467])
- Dec SSL is robust to heterogeneous distributed dataset (ICLR'23 [1258])
- Predict client training time to guide dynamic epoch (WCMC'22 [1581])

1.2 Communication Efficiency

Communication efficiency is important in federated learning

- Structured updates and sketched updates (arXiv'16 [589])
- MOCHA (NeurIPS'17 [1099]): Systems-aware via multi-task learning
- Matcha (ICC'19 [1249]): Speed up via matching decomposition sampling
- AFL (arXiv'19 [326]): Select with probability conditioned rather uniformity
- FedAsync (arXiv'19 [1411]) has near-linear convergence to a global optimum
- LotteryFL (arXiv'20 [630]): Exploit lottery ticket hypothesis
- Topology to speed up by maximizing system throughput (NeurIPS'20 [862])

HeteroFL (ICLR'21 [218]): Local model with diff computation complexity
 Attention-based and dynamic fraction for client selection (IJCAIW'21 [147])
 Compress gradient with error-compensated compression (AAAI'21 [306])
 FedPara (ICLR'22 [483]): Low-rank hadamard product to re-parameterize
 FedAGM (arXiv'22 [568]): Global gradient guides client updates
 Generalization FL: out-of-sample gap and participation gap (ICLR'22 [1561])
 FedNew (ICML'22 [250]): Not transmit Hessian from clients to server
 FedDST (AAAI'22 [46]) Dynamic Sparse train sub-networks in FL
 Select device, allocate transmit power, receive beamforming (SCIS'22 [313])
 FedCAMS (ICML'22 [1303]) Variant of FedAdam feature max stabilization
 FedSI (SCIS'23 [1672]): Communication-efficient FL continual learning
 Keep PLMs frozen and transfer lightweight adapter (ACL'23 [786])
 DepthFL (ICLR'23 [573]): Different depths with mutual self-distillation
 FEDSPEED (ICLR'23 [1128]): Prox-correction reduces the prox bias
 FedGradMP (arXiv'23 [499]): Gradient matching for sparsity constraint
 Fed-LAMB (UAI'23 [554]): Layer-wise and Dimension-wise adaptivity
 HCSFed (UAI'23 [1100]): Compressed gradient and clustering-based select
 FedPart (NeurIPS'24 [1238]): [Selecting and training partial networks](#)

1.3 Data Heterogeneity

1.3.1 Client Calibration

Shared Global Network

Utilize the global model to guide the local updating.

FedAvg (AISTATS'17 [868]): Average parameter to realize communication
 FedProx (MLSys'20 [678]): Introduce proximal term (global parameter)
 FedCurv (NeurIPS'19 [1089]): Leverage hessian matrix (global parameter)
 SCAFFOLD (ICML'20 [555]): Correct the update direction (gradient)
 FLMoE (arXiv'20 [1577]): Using a mixture of experts
[MOON \(CVPR'21 \[668\]\): Align feature-level agreement with global model](#)
 FedUFO (ICCV'21 [1614]): Unify feature learning and optim objective
 FedDyn (ICLR'21 [2]): Based on Dynamic Regularization
 Implicit Gradient Alignment in Federated Learning (AAAI'22 [198])
 FedDC (CVPR'22 [307]): An auxiliary local drift variable to track the gap
[FedNTD \(NeurIPS'22 \[614\]\): Preserve the global view for not-true classes](#)
 CLIMB (ICLR'22 [1065]): FL Agnostic with Class Imbalance
 FedPVR (CVPR'23 [633]): Partial variance reduction on last layers
 FedSeg (CVPR'23 [877]): Modified cross-entropy loss and pixel-level CL
 GradMA (CVPR'23 [823]): Correct server and worker-side via GEM
 FedBR (ICML'23 [368]): Global label agnostic pseudo for feat and classifier
 Fed-GraB (NeurIPS'23 [1406]): Self-adjust and closed-loop gradient re-balance
 FedGuCci (arXiv'24 [714]): Global and anchor models for group connectivity
 FedNLR (SIGKDD'24 [1244]): Enhance the neurons learning bound to local classes on local data knowledge

TAKFL (NeurIPS’24 [889]): Transfer knowledge from each prototype’s ensembles

Extra Network Architecture

Utilize the extra network architecture to guide the local updating.

FedGKT (NeurIPS’20 [405]): Group knowledge transfer for large CNNs

FedMLB (ICML’22 [569]): Align to multiple auxiliary branches

FedCGAN (IJCAI’22 [1385]) Conditional GAN in FL

ADCOL (ICML’23 [669]): Introduce discriminator module

DaFKD (CVPR’23 [1235]): GAN evaluates client importance to distill sample

Global Statistic Information

Utilize the global class information to guide the local updating.

FedProc (arXiv’21 [894]): Global prototypes with InfoNCE

Local imbalance and global imbalance in FL (AAAI’21 [1256])

CCVR (NeurIPS’21 [824]): Approximated GMM to calibrate classifier

HarmoFL (AAAI’22 [514]): Normalize amplitude and weight-perturbation

FedTFI (SCIS’22 [956]): Random exchange local feature maps for IIoT

FedFA (ICLR’23 [1719]): Federated feature aug via Gaussian distribution

[FPL \(CVPR’23 \[468\]\): Rethink FL under domain shift: a prototype view](#)

FedIIC (MICCAI’23 [1372]): Intra and inter contrast difficulty logit adjust

FedCSD (arXiv’23 [1471]): Prototype similar distill and adaptive mask filter

FedImpro (ICLR’24 [1154]): Construct similar conditional distribution

FedFA+ (PAMI’24 [1725]): Exploits feature statistics and explicit alignment

FOOGD (NeurIPS’24 [736]): Estimate score model and enhance invariant rep.

Self Regularization

Utilize the self-regularization solution to calibrate the local direction

FedRS (KDD’21 [688]): Restricted Softmax to limit missing classes’ weights

Fed² (KDD’21 [1552]): A firm structure-feature alignment

FedAlign (CVPR’22 [872]): Local learning matters in data heter FL

FedSAM (ICML’22 [993]): Local generality via SAM

FedLC (ICML’22 [1603]): Calibrate logits based on probability of each class

[FedDecorr \(ICLR’23 \[1080\]\): Regularize Frobenius norm of correlation matrix](#)

Client-specific scalars for local and global calibration (ICML’24 [959])

Federated Data Sharing

Utilize the Federated data to support optimization

Share a subset of shared datasets from the cloud server(arXiv’18 [1690])

DS-FL (TMC’21 [490]): Share unlabeled data

DC-Adam (CS'21 [1166]): Warm-up operation

FEDAUX (TNNLS'21 [1036]): Pretrain and distill unlabeled auxiliary data

Federated Data Augmentation

Utilize the Federated data augmentation strategy to provide diverse local data

FAug (arXiv'18 [498]): GAN generator supplements target tags

Astraea (ICCD'19 [240]): Augment based on the global data distribution

XorMixUp (arXiv'20 [1087]): XorMixup hybrid data enhancement

FedMix (ICLR'21 [1547]): Leverage Mixup without leaking privacy

Fed-ZDAC (CVPRW'21 [391]): Fair fl with zero-shot data augmentation

FEDGEN (ICML'21 [1761]): Data-free knowledge distillation via generator

FedLESAM (ICML'24 [267]): Locally estimate the direction of global perturbation

Federated Data Selection

Utilize the Federated data selection to choose the local data

MinCost (TPDS'20 [1219]): Average cost minimization and greedy algorithm

FAVOR (INFOCOM'20 [1234]): Intelligently chooses clients via q-learning

FEDSS (ICC'20 [61]): Dynamically select samples according to available size

FedACS (IWQOS'21 [1319]): Clients with small heterogeneity via Thompson

FedAgg (IWQOS'21 [1452]): Select devices to converge via model splitting

DQS (ICN'21 [1136]): Prioritize reliable devices with rich and diverse datasets

[FedCor \(CVPR'22 \[1148\]\): Model the client correlations with a GP](#)

Select client with higher loss yields faster convergence (AISTATS'22 [163])

Safe (TII'22 [1454]): Filtering poisoned data by clustering

1.3.2 Server Calibration

Global Optimization

Conduct the optimization on the server side

Synthesize datasets with a continuous range of identicalness (arXiv'19 [437])

IDA (MICCAIW'20 [1532]): Calculates weights via statistical meta-information

FedOPT (ICLR'21 [1007]): Adaptive optim for heterogeneity and efficiency

FedSAE (IJCNN'21 [656]): Workload prediction adjusts equipment tasks

FedFTG (CVPR'22 [1615]): Fine-tuning Global Model via Data-Free KD

CReFF (IJCAI'22 [1046]): Classifier re-training with federated features

FedAda (WWW'22 [1596]): Multi-objective finds optimal iterations

FedDM (CVPR'23 [1429]): Local learns synthetic data for server to update

DYNAFED (CVPR'23 [974]): Synthesize from dynamics of global trajectory.

CLIP2FL (AAAI'24 [1077]): CLIP guides features generation via prototype

SAFARI (ICML'24 [1485]): Deploy an auxiliary dataset at server

Global Reweighting

How can the aggregation weights be assigned to generate a global DNN model?

BNPFL (ICML'19 [1573]): Bayesian for probabilistic matching

SPAHM (NeurIPS'19 [1572]): Bayesian non-parameter match clients weights

FedNova (NeurIPS'20 [1247]): Normalized average for objective inconsistency

FedMA (ICLR'20 [1243]): Layer-wise Bayesian nonparametric aggregation

OT (NeurIPS'20 [1095]): Layer-wise alignment via optimal transport

FRCS (AAAI'21 [900]): Game theory and Shapley value for client selection

FEDPA (ICLR'21 [5]): Federated learning via posterior averaging

FEDBE (ICLR'21 [106]): Aggregate via Bayesian model Ensemble

FEDDUALAVG (ICML'21 [1562]): Perform averaging in the dual space

Auto-FedAvg (arXiv'21 [1397]): Learnable weight optimized on clients data

DivFL (ICLR'21 [36]): Small diverse clients with representative gradient

FedSim (NC'22 [927]): Clients with similar gradients are clustered

Reduce updates to more sensitive parameters via Elastic (CVPR'23 [216])

FEDLAW (ICML'23 [715]): Global weight shrinks and is smaller than 1

FedGMA (ICPP'23 [443]): Handle advanced models and memory bloating

ShapleyFL (SIGKDD'23 [1123]): Shapley adaptive weighting

PI-Fed (IoTJ'24 [1553]): Gradient regularization, parameter-importance aggregation and soft-masking

FedConcat (AAAI'24 [219]): Concatenate local models via clustering

1.3.3 Federated Domain Adaptation

Federated domain adaptation fields.

A domain expert model and a private gating mechanism (NeurIPS'19 [970])

FADA (ICLR'20 [964]): Focus on domain adaption in federated learning

COPA (ICCV'21 [1360]): Ensemble of domain-specific classifiers

KD3A (ICML'21 [281]): Knowledge Vote for high-quality domain consensus

MCC-DA (TCSVT'22 [1339]): Collaborative contrastive alignment

GMFedDA (TNNLS'22 [1579]): Pretrain and gradient matching

MCKD (ICASSP'23 [915]): Error remove + knowledge filter for mutual KD

IRCMSDA (ICASSP'23 [1336]): Instance relation consistency

AEGR (ICME'23 [643]): Adversarial Examples filter out errors

Co-MDA (TCSVT'23 [784]): Distill via domain att and remove confirm bias

FedICON (NeurIPS'23 [1141]): Global invariance and test consistency

1.3.4 Federated Domain Generalization

Federated domain generalization fields.

FedDG (CVPR'21 [769]): Federated DG on unseen domain

COPA (ICCV'21 [1360]): Ensemble of domain-specific classifiers

FedADG (arXiv'21 [1612]): Federated adversarial to align distributions

FedSR (NeurIPS'22 [906]): Simple l_2 and conditional mutual: CMI and L2R

IOP-FL (TMI'23 [515]): Test-time routing with consistency and shape loss
 CCST (WACV'23 [108]): Augment data via other source style, like FedFA
 FedGA (CVPR'23 [1628]): Measure fairness by generalization gaps variance
 CSAC (TKDE'23 [1563]): Semantic aggregate and calibrate in layer wise
 MCKD (ICASSP'23 [915]): Error remove + knowledge filter for mutual KD
 FedINB (ICLR'23 [1728]): Federated learning with iterative naive barycenter
 FedFusion (CVPR'23 [239]): Regard aggregation as virtual components
 FedADG (IOTJ'23 [1613]): Federated adversarial learning for domain-invariance
 FedPD (ICCV'23[1480]): Disentangle para and global divide conquer

1.3.5 Personalized Federation

Local model with personalization operation

FEDPER (arXiv'19 [21]): Base layers and personalized layers
 Divide model into federated and private parameters (arXiv'19 [58])
 FL personalization via model agnostic meta learning (arxiv'19 [521])
 Mix local and global models with optimal mixing parameter (arXiv'20 [215])
 share generic feature and keep a local private classifier (arXiv'20 [1491])
 MAFL: An initial shared model for current or new users (NeurIPS'20 [257])
 pFedME (NeurIPS'20 [1135]): Moreau envelopes with bi-level optimization
 LG-FEDAVG (NeurIPS'20 [726]): Global model + Local representation
 FML (arXiv'20 [1061]): Leverage shared model to conduct mutual learning
 FEDRECON (NeurIPS'21 [1096]): Model-agnostic for scale learning
 FedAMP (AAAI'21 [472]): Similar clients to collaborate more
 pFedHN (ICML'21 [1044]): Central hypernetwork outputs personal model
 FedFomo (ICLR'21 [1618]): Personalized weighted combination of others
 PGFL (arXiv'21 [879]): Prototypes and attention for aggregation
 FedBN (ICLR'21 [683]): Except Batch Normalization module
 FedCG (CVPRW'21 [67]): Cluster-driven graph FL over multiple domains
 WAFFLe (Access'22 [392]): Weight anonymized factorization for fl
 FedProto (AAAI'22 [1143]): Global prototypes with L1 regularization
 CGPFL (IJCAI'22 [1151]): Handle context-level heterogeneity in FL
 Personalize could benefit with small personal parameters (ICML'22 [976])
 Alternating update outperforms the simultaneous update (ICML'22 [976])
 kNN-Per (ICML'22 [861]): Interpolate global with local k-nearest neighbors
 pFedLA (CVPR'22 [846]): Discern the importance of each layer
 Channel Decoupling for Personalization Federated Learning (CVPR'22 [1064])
 FED-ROD (ICLR'22 [107]): Decouple generic and personalized with two

tasks

SplitMix (ICLR'22 [423]): In-situ customization and adversarial robustness
 FedBABU (ICLR'22 [920]): Only updates model body during FL
 SFL (AAAI'22 [98]): Client-wise relations and Graph- based topology
 Layer-wise model aggregation for scalable FL (CVPR'22 [846])
 FedReg (ICLR'22 [1432]): Regularize with generated pseudo data
 FedPCL (NeurIPS'22 [1144]): Pretrained backbones and contrast prototypes
 FedSM (CVPR'22 [1430]): Model selector to decide the closest model

APPLE (IJCAI'22 [822]): Adaptively learning via directed relationship
 On the convergence of clustered federated learning (arXiv'22 [840])
 Specificity-Preserving FL for MR Reconstruction (TMI'22 [280])
 FedNH (AAAI'23 [192]): Uniform prototypes + infuse semantics
 Adapt the ETF structure in FL training (ICCV'23 [716])
 FedTHE (ICLR'23 [511]): Test-time head ensemble and feature alignment
 FedPAC (ICLR'23 [1443]): Feature align and learnable classifier combine
 FedALA (AAAI'23 [1599]): Adaptive aggregates in element-wise
 FedDAR (ICLR'23 [1703]): Shared backbone and domain-wise heads
 FedHKD (ICLR'23 [102]): Means of local features and soft predictions
 DynamicPFL (NeurIPS'23 [1503]): Dynamic PFL with adaptive DP
 PGFed (ICCV'23 [821]): Personalize client global objective via loss aggregate
 GPFL (ICCV'23[1598]): Learn global and personalized feature information
 Learn2pFed (arXiv'24 [830]): Dynamically determine local parameters
 FedLoGe (ICLR'24 [1405]): Static Sparse ETF and feature realignment
 FedTGP (AAAI'24 [1604]): Adaptive-margin contrast trainable global proto
 (CVPR'24 [1504]):
 Balancing Similarity and Complementarity for FL (ICML'24 [1470])
 PERADA (CVPR'24 [1408]): Global/personalized adapter and knowledge
 distillation
 DualFed (MM '24 [1739]): Insert personalized projection between encoder
 and classifier

Model Heterogeneity

The model with different architecture

Cronus (arXiv'19 [84]): Public dataset (soft labels) is used with local data
 KT-pFL (NeurIPS'21 [1597]): Coefficient matrix for local predictions
 FedMD (NeurIPS'19 [638]): Rely on related public data to distillation
 RCFL (KDDW'20 [356]): Overlapping labels with weight aggregation
 FedDF (NeurIPS'20 [745]): Ensemble distillation for model fusion
 FedAD (ICCV'21 [328]): Ensemble attention distill on unlabeled public data
 FEDMD-NFDP (IJCAI'21 [1119]): FedMD in Noise-Free Differential Pri-
 vacy
 FedHeNN (ICML'22 [855]): CKA for architecture agnostic federated learning
 Data heterogeneity limits knowledge distillation (ICLR'22 [3])
 FCCL (CVPR'22 [464]): Align Cross-Correlation Matrix with public data
 FSMAFL (ACMMM'22 [465]): Few-shot model agnostic federated learning
 FCCLPLUS (TPAMI'23 [466]): Cross-Correlation and Instance Similarity

1.3.6 Cluster

Clustered FL groups clients with the same data distribution into a cluster

Group patients into clusters based on their electronic medical records ([766])
 CFL (TNNLS'20 [1037]): Split based on cosine similarity of client gradients
 FL+HC (IJCNN'20 [53]): Measure similarity of local updates to global

IFCA (NeurIPS'20 [321]): Alternately estimate the cluster identities
HYPCLUSTER (arXiv'20 [859]): Assign to cluster with lowest local loss
FedGroup (ISPA'21 [241]): Cosine similarity on decomposed gradient
Group patients into clusters based on imaging modality (ITJ'21 [11])
FedSoft (AAAI'22 [1026]): Soft clustered FL with proximal local updating
MFCL (WWW'22 [801]): Multi-center aggregation via parameter
FedRC (ICML'24 [369]): Cluster principle by a bi-level optimization

1.3.7 Unsupervised learning

Federated learning meets unsupervised learning

FURL (arXiv'20 [1588]): Federated unsupervised representation learning.
FedSSL (ICLR'22 [1766]): FL meets Self-Supervised learning.
Orchestra (ICML'22 [815]): Cluster-based SSL exploits federated hierarchy

1.3.8 Semi-Supervised Learning

Semi-supervised learning faces federated learning

Semi-supervised Vertical FL with MultiView Training (IJCAIW'20 [551])
FedMatch (ICLR'21 [500]): Inter-client consistency and disjoint parameters
Private Semi-Supervised Federated Learning (IJCAI'22 [261])
CBAFed (CVPR'23 [664]): Fixed pseudo handles catastrophic forgetting

1.3.9 Noise Learning

Federated learning with noise learning.

FOCUS (IJCAIW'20 [145]): Credibility-weighting with mutual CE
CoMT (AAAI'20 [387]): Craft instances and subset-select the training set
RHFL (CVPR'22 [274]): Robust fl with noisy and heterogeneous clients
FedLSR (CIKM'22 [520]): Avoid memorize noise and output consistency
Select via local image quality and computation capability (Access'20 [1524])

1.3.10 Multi Modality

FL faces multi-modality problem

FL for vision-and-language grounding problems (AAAI'20 [757])
Cross-modal federated human activity recognition (AAAI'22 [1505])
CreamFL (ICLR'23 [1557]): Global-local cross-modal ensemble
FedCMI ([266]): Transfer the global dominant knowledge

1.3.11 Continual Learning

FL faces continual learning scenario

FL continual learning via weighted inter-client transfer (ICML'21 [1546])
CFed (IJCAI'22 [851]): Continual FL based on knowledge distillation
FedSpace (CVPR'23 [1070]): Asynchronous federated continual learning

1.4 Fairness

Collaborative fairness and performance fairness in federated learning.

1.4.1 Performance Fairness

Involve diverse datasets and improve individual prediction accuracy

- AFL (ICML'19 [887]): Maximize the performance of worst performing client
- qFFL (ICLR'20 [679]): Emphasize the clients with larger loss
- FADE (KDD'21 [425]): Yield biased and unfair models for minority groups
- FCFL (NeurIPS'21 [187]): Gradient-based constrained multi-objective
- Ditto (ICML'21 [676]): A lightweight personalization for global fl objective
- FedMGDA+ (TNSE'20 [447])
- FedFV (IJCAI'21 [1314]): Detect gradient conflicts via cosine
- FedMDFG (AAAI'23 [932]): Unfair direction and an improper step size
- FedCE (CVPR'23 [513]): Measure contribution from gradient and data space
- FairFed (AAAI'23 [256]): Fairness-aware aggregation for group fairness
- FedBEAL (CVPR'23 [1459]): Eliminate bias aug for bias-conflict samples
- GIFAIR (IJDS'23 [1568]): Penalize the loss spread to fair solutions
- Scaff-PD (arXiv'23 [1559]): Optimize distributionally robust objectives
- FedEBA+ ([1254]): Entropy-Based Aggregation plus alignment update ([139])

1.4.2 Collaboration Fairness

Treat participants fairly based on their contributions towards FL system.

- Group instance deletion and group Shapley values ([1227])
- Contract theory-based incentive mechanism (APWCS'19 [548])
- FedCI(BigData'19 [1102]): Shapley by val accuracy of intermediate models
- Local computation and communication estimate contribution (TOJ'19 [546])
- Define Contribution Index with three approximations (FL'20 [1333])
- Consortium blockchain for reputation management (WC'20 [549])
- CFLL (FL'20 [832]): Reputations enforces client towards different models
- Stackelberg analyzes the CPU and budget allocation for clients ([1035])
- Data value estimator of the gradients via RL (ICASSP'21 [1679])
- Incentive between BS and mobile users as an auction game (TWC'21 [610])
- RRFL (ICMLW'21 [1455]) Reputation for col fairness and adv robustness
- CGSV (NeurIPS'21 [1456]): Shapley by COS between local and global update
- FedFAIM (TBD'22 [1086]): Erase low-quality gradients via loss measurement
- Beta distribution to measures the local user credibility (IoTJ'22 [1105])
- RRCM (IJCAIW'22 [1607]): Reward-punish and cost-interest compensate
- HFFL ([1602]): Reward agents via pre-negotiated contribution levels
- SDRCGM (AAAI'22 [1159]): Synthetic data for parties as rewards
- Payoff and model rewards in Shapley-fair (NeurIPS'22 [909])

Enabling competitors to collaborate in federated learning (TBD'22 [1382])

1.5 Robustness

1.5.1 Byzantine Attack

Byzantine Data-Level Poisoning in FL

Papers focus on Byzantine Data-Level Poisoning

- Adversarial machine learning (ACM Workshop'11 [456])
- Label flipping via gradient ascent (ICML'12 [48])
- Symmetry Flipping (NeurIPS'15 [1198])
- Badnets: Identify vulnerabilities in ML model supply chain (arXiv'17 [350])
- Pair Flipping (NeurIPS'18 [378])
- Asynchronous Byzantine machine learning (ICML'18 [197])

Byzantine Model-Level Poisoning in FL

Papers focus on Byzantine Model-Level Poisoning

- Norm clipping and “weak” differential privacy (NeurIPS'19 [1130])
- LIE (NeurIPS'19 [41]): A perturbation range of malicious client parameter
- Consider adversaries to only cause targeted poisoning (ICML'19 [45])
- IPM (UAI'20 [1410]) Inner production manipulation attacks
- Fang (USNEIX'20 [272]): Manipulate model via compromising some clients
- Defend against Targeted Model-Poisoning Attacks on FL (arXiv'23 [980])
- SignGuard (ICDCS'22 [1441]): Norm thresholding and sign clustering filter.
- MPAF (CVPR'22 [71]): Drag global towards an attacker with low accuracy
- ClusterAttack (AAAI'23 [1560]): Poisonous gradients lead dense clusters

1.5.2 Backdoor Attack

Federated learning with backdoor attack

- How to backdoor federated learning (AISTATS'20 [27])
- DBA (ICLR'20 [1409]): Decompose global backdoor into separate ones
- Dynamic backdoor attacks against federated learning (arXiv'20 [450])
- FL is susceptible to edge-case backdoors (NeurIPS'20 [1240])
- Neurotoxin(ICML'22 [1667]): Project adversarial gradient onto subspace unused by benign
- Learning to backdoor federated learning (ICLRW'23 [647])
- F3BA (AAAI'23 [273]): Flip some unimportant parameters in main task
- CerP (AAAI'23[836]): Jointly tune trigger and control poison changes
- RLBackdoor (ICLRW'23 [647]): Reinforcement learning-based backdoor
- Chameleon (ICML'23 [193]): Pull benign and poisoned with same labels
- IBA (NeurIPS'23 [910]): Involves trigger generating and backdoor injection ([1590])
- Backdoor threats from compromised foundation to FL (NeurIPSW'23 [686])

FCBA (AAAI'24 [776]): Combinatorics theory to design trigger strategies
BCLayer (ICLR'24 [1763]): Existence of backdoor-critical (BC) layers

1.5.3 Malicious Defense

Focus on Malicious Defense scenarios

Distance Base Algorithms

Distance base Byzantine Defense

Krum (NeurIPS'17 [50]): Select one is similar to others as the global model.
FoolsGold (arXiv'18 [301]): Identify sybils based on client updates diversity.
FABA (AAAI'19 [1395]): Remove the outliers in the uploaded gradients
AFA (arXiv'19 [898]): Cosine remove and Hidden Markov estimation
A secure federated transfer learning framework (TS'20 [793])
Robust FL with attack-adaptive aggregation (IJCAIW'21 [1214])
DnC (NDSS'21 [1056]): Spectral analysis and product to remove large one
MAB-RFL (IJCAI'22 [1216]): Robust aggregate with adaptive client select
DeFL (AAAI'23 [1468]): Fine-grain difference via federated gradient norm
FLCert ([74]): Divide clients into groups with majority vote
Dim-Krum (EMNLP'22[1669]): Calculate distances on small ratio of dims
FedCPA (ICCV'23 [382]): Benign with similar critical parameters
Multi-metrics adaptively identifies backdoors (ICCV'23 [459])
(SP'23 [603])
FreqFed (NDSS'24 [289]): Transform local weights into the frequency domain
DFLDual (CVPR'24 [1122]): Dual-domain client cluster and trust bootstrap
([38])

Statistics Distribution Schemes

Statistics distribution Byzantine Defense

Bulyan (ICML'18 [357]): Combine Krum and a variant of trimmed mean
Median (ICML'18 [1537]): *Median* and *trimmed-mean* via gradient descent
AAA (IJCAIW'21 [1215]): Robust attack-adaptive aggregation with attention
RFA (TSP'22 [975]): Based on geometric median and smoothed Weiszfeld
FLDetector ([1659]): Check received and predicted updates consistency
FedRecover (SP'23 [73]): Server utilizes historical information to recover

Proxy Dataset Algorithms

Proxy dataset Byzantine Defense

FedRAD (arXiv'21 [1109]): Detect adversaries via the output predictions
BatFL (IWQO'21 [1389]): Evaluate the contribution via Shapley Value
FLTrust (NDSS'21 [70]): ReLU-clipped cosine similarity with proxy dataset
Sageflow (NeurIPS'21 [947]): Entropy-based filter and loss-weighted average
SmartFL (arXiv'22 [1420]): Aggregate via server-collected proxy data

FedGame (NeurIPS'23 [504]): Compute a genuine score via minmax game
SDEA (ICML'24 [462]): Self-Driven Entropy Aggregation
Maliciousness score detection and evil number estimator (ICML'24 [550])

Target Optimization based

Optimize a different objective function

FedPruning (arXiv'20 [1357]): Remove redundant and adjust extreme weight
RSA (AAAI'19 [660]): Norm regularization and stochastic subgradient
RLR (AAAI'21 [925]): Adjust server lr, per dimension and per round
ProvableFL (AAAI'21 [72]): Multiple global models with majority voting
RB-TM (NeurIPS'21 [249]): Reliable broadcast and witness to collect others
SparseFed (ICML'22 [933]): Global top-k sparsification and gradient clipping

Client Side Defense

Consider client self-defense against

CRFL (ICML'21 [1407]): Clip and smooth control the global smoothness
[LeadFL \(ICML'23 \[1731\]\)](#): [Hessian matrix of local gradient is nullified](#)
FLIP (ICLR'23 [1609]): Adversarial trains local on generated backdoor
Unlearn information unrelated to backdoor patterns (arXiv'23 [6])
FedDefender (SIGKDD'23 [951]): Client-side meta update and global KD
SDFC (ECCV'24 [467]):

1.5.4 Free-riders Attack

Free-riders in Federated Learning: Attacks and Defenses ([739])
(NeurIPS'21 [1762])
(AISTATS'21 [294])
(ISCC'22 [1245])

1.5.5 Free-rider Defense

(IoTJ'19 [547])
(IoTJ'21 [1369])
(IoTJ'23 [1217])
(JMLR'23 [628])
(IoTJ'23 [1458])
(IS'24 [109])

1.5.6 Noise Learning

Federated learning with noise learning

FedCorr (CVPR'22 [1439]) Multi-Stage FL for Label Noise Correction
FedNoRo (IJCAI'23 [1371]): Global data is class-imbalanced and label noise
FedNed (AAAI'24 [812]): Identify noisy via uncertainty and negative KD

1.5.7 Adversarial Learning

Federated learning with adversarial training.

FedRN (CIKM'22 [577]): Retrieve k-reliable neighbors and identify examples

DBFAT (AAAI'23 [1601]): Local re-weight and global regularization

FedBRN (AAAI'23 [424]): Propagate adv robustness from rich to poor

1.6 Large Model

Federated learning meets the large language model.

Generative pre-trained model-assisted federated learning (arXiv'23 [1637])

Foundation Model Meets FL: Motivations, Challenges, Directions (arXiv'23[1765])

1.6.1 Large Language Model

LP-FL (arXiv'23 [510]): Average Lora and assign soft labels

Personalization and Robustness Trade-off in FedPrompt (arXiv'23 [178])

FedPrompt (ICASSP'23 [1676]): Freeze PLMs and aggregate prompts

FedBPT (ICML'24 [1118]): Train optimal prompts and gradient-free optim

FedSA (arXiv'24 [363]): Share A for general and keep B for client-specific

DEeR ([1746]): Aggregation deviation and noise amplification for LoRA

DoFIT ([1431]): Domain-aware Fed Instruction Tuning

[FLoRA \(NeurIPS'24 \[1315\]\): Stacking-based LoRA aggregation](#)

FedCoLLM (arXiv'24 [263]): Co-tune Large-LMs and Small-LMs

1.6.2 VQA and Time Series Forecasting

VQA and Time Series Forecasting

STP (IJCAI'23 [122]): Federated spatial-temporal prompt for weather forecasting

FedDAT (AAAI'24 [104]): Dual-Adapter with mutual knowledge distillation

1.6.3 CLIP

CLIP based FL

FedAPT ([1110]): Cross-domain federated adaptive prompt tuning

PromptFL ([365]): Cooperatively train shared soft text prompts

FedCLIP (arXiv'23 [807]): Attention-based adapter for CLIP

pFedPG (ICCV'23 [1483]): Client visual prompt generator

ATPFL (UbiComp'23 [1693]): Text and Visual contrast with global prompt

Visual prompt improves the privacy accuracy under DP (ICCV'23 [703])

Compose prompts together with modified attentions (CVPR'23 [52])

Fed-DPT (arXiv'23 [1324]): Visual and textual prompt tuning

pFedPrompt (WWW'23 [364]): Personalized att and linguistic consensus

FedTPG (ICLR'24 [987]): Learn a unified prompt generation network across

FedOTP (CVPR'24 [644]): Global and local prompts cooperation via OT

FedAPT (AAAI'24 [1111]): Domain-specific keys to generate specific prompts
DiPrompt (CVPR'24 [31]): Disentangled prompt: global and domain
SGPT (CVPR'24 [210]): Shared and group prompt tuning for both common and group-specific features

1.6.4 ViT

ViT based FL

FedPR (CVPR'23 [279])
pFedPT (arXiv'23[642]): Personalized visual prompt and shared ViT
FedPrefix (ICCV'23 [1113]): Partially personalize ViT
FedHPL (arXiv'24 [849]): Local prompt tuning and global tuning

1.7 To Classification

(AAAI'21 [901])
(TNNLS'23 [141])

1.8 Differential Privacy

Secure discussion in federated learning

Unintended features leakage exposes to powerful inference SSP'19 [870]
Reconstruct images from parameter gradients (NeurIPS'20 [319])
FPCA (NeurIPS'20 [334]): Combine FL and differential privacy
Private federated learning with laplacian smoothing (NeurIPS'20 [731])
Soteria (CVPR'21 [1116]) Perturb representation to decrease reconstruction
Clip FL: Convergence and Client-Level Differential Privacy (ICML'22 [1641])
Obtain Private Data in FL with Modified Models (ICLR'22 [293])
Privacy Leakage of Adversarial Training Models (CVPRW'22 [1595])
Clipped FedAvg perform well because similar client updates (ICML'22 [1642])
Security of FL: Attacks, defensive mechanisms, and challenge ([44])
Federated learning attack surface: taxonomy, cyber defences, challenges, and future directions (AIR'22 [981])
Differentially Private Federated Learning with SAM (CVPR'23 [1081])
Sparsity for resource problem in linear layer leakage (CVPR'23 [1680])
Class-wise recovery via paired-logits inversion attack (CVPR'23 [1137])
iLRG (ICLR'23 [841]): Instance-wise label restoration from gradients
LTI (UAI'23 [1374]): Auxiliary dataset trains a model to invert gradient

1.9 Novel direction

I fail to classify into existing major research fields

Lazily aggregated quantized gradient for FL (TPAMI'20 [1115])
Federated deep learning via neural architecture search (CVPRW'20 [404])

Federated Bayesian optimization via Thompson sampling (NeurIPS'20 [196])
Transformers are more robust to heterogeneous FL (CVPR'22 [991])
Federated hyper-parameter tuning (NeurIPS'21 [563])
Communication-efficient random for multi-kernel online fl (TPAMI'21 [427])
Heterogeneous Ensemble KD for Large Models in FL (IJCAI'22 [162])
MaKEr (IJCAI'22 [119]): Meta-learning knowledge extrapolation for KG
Interleave aggregation and permutation for small data (ICLR'23 [542])

1.10 Application

Applications in federated learning Federated Learning for Mobile Keyboard Prediction (arXiv'18 [397])

Federated Prediction Model for Stroke Prevention (arxiv'20 [535])
Federated Person Re-identification via Benchmark Analysis (MM'20 [1767])
Federated Learning with Humor Recognition (IJCAI'20 [367])
Fedvision: An Online Visual Object Detection Platform (AAAI'20 [791])
FedDG: FL Domain Generalization on Medical Image (CVPR'21 [769])
FedReID (AAAI'21 [1361]): Multi-Domain Labels for Person ReID
Jupiter (SCIS'21 [1425]): A modern FL platform for regional medical care
FedMAT (IJCAI'22 [1060]): Cross-individual human activity recognition
FedScale (ICML'22 [605]): Scalable and reproducible benchmark for reality
FL for rare cancer boundary detection (Nature Communication [955])

1.11 Survey

Survey about federated learning

Federated learning with Non-IID Data (arXiv'18 [1690])
Federated machine learning: Concept and applications (TIST'19 [1493])
3 Approaches for Personalization with Applications to FL (arXiv'20 [858])
Fair and privacy-preserving federated deep models (TPDS'20 [835])
Survey of personalization techniques for federated learning (WordS4'20 [598])
FL: Challenges, Methods, and Future Directions (SPM'20 [677])
Personalized fl for intelligent IoT applications (OJ-COMS'20 [1373])
A federated transfer learning for wearable healthcare (TS'20 [142])
Federated transfer learning: Concept and applications (IA'21 [1029])
Threats to federated learning: A survey (arXiv'20 [834])
Fair and differentially private decentralised deep learning (TDSC'20 [831])
Asynchronous federated learning on heterogeneous devices (arXiv'21 [1433])
Federated ML: Survey, classification, criteria and directions (CST'21 [1212])
Federated Learning: Issues in Medical Application (FDSE'21 [1544])
A Survey on Federated Learning Systems (TKDE'21 [670])
A survey on security and privacy of federated learning (FGCS'21 [891])
Advances and open problems in federated learning (FTML'21 [540])
Federated learning on non-IID data: A survey (NC'21 [1740])

Towards Personalized Federated Learning (TNNLS'22 [1138])
FL A taxonomy, review, and future directions (CSUR'21 [1542])
FL on non-iid data silos: An experimental study (ICDE'22 [665])
Federated Learning for Smart Healthcare: A Survey (CSUR'22 [908])
Survey on solving non-IID data in FL (FGCS'22 [848])
Privacy and Robustness in FL: Attacks and Defenses (TNNLS'22 [833])
Challenges and approaches for byzantine attacks in FL (TrustCom'22 [1076])
A critical evaluation of poisoning attacks on FL (IEEE SP'22 [1057])
FL on Non-IID Data Silos: An Experimental Study (TKDE'22 [666])
Vertical federated learning (arXiv'22 [794])
VFL: Challenges, methodologies and experiments (arXiv'22 [1329])
Study of Byzantine-Robust Aggregation in FL (TBD'23 [672])
Heterogeneous FL: SOTAs and Research Challenges (CSUR'23 [1526])
Federated Domain Generalization: A Survey (arXiv'23 [706])
Scale, privacy, and hetero introduce noise in FL tuning (MLsys'23 [604])
Fairness and Privacy-Preserving in FL (arXiv'23 [1000])
Towards fairness-aware federated learning (TNNLS'23 [1083])
FL threats: Concepts, taxonomy on attacks and defenses (IF'23 [1017])
A robust analysis of adversarial attacks on FL (CSI'23 [903])
Federated Learning without Full Labels: A Survey (arXiv'23 [527])
A Survey of What to Share in Federated Learning (arXiv'23 [1052])
Privacy and fairness in FL: on the Perspective of Trade-off (CSUR'23 [105])
FL with New Knowledge: Fundamentals, Advances, Futures (arXiv'24 [1259])
Federated GNN: Overview, Techniques, and Challenges (TNNLS'24 [772])
Adversarial Attacks Impact on Federation: A Survey (TPAMI'23 [602])
[Federation for Generalization, Robustness, Fairness \(TPAMI'24 \[469\]\)](#)

Chapter 2

Multimodal Learning

Multimodal learning focuses on integrating different senses.

2.1 Survey

Some survey about multimodal learning

- Development of embodied cognition: six lessons from babies (AL'05 [1098])
- Multiscale Benchmarks for Multimodal Representation (NeurIPSW'21 [728])
- Learning in Audio-visual Context (arXiv'22 [1337])
- Multimodal learning with transformers: A survey (PAMI'23 [1448])
- A systematic survey of prompting methods in NLP (ACMCS'23 [768])
- Multimodal foundation: specialists to general assistants (arXiv'23 [637])
- A Standardized Toolkit for Multimodal Deep Learning (JMLR'23 [727])
- A survey of prompt on the vision-language foundation (arXiv'23 [345])
- A Survey on Multimodal Large Language Models (arXiv'23 [1540])
- LLM hall: principles, taxonomy, challenges, questions (ACM TIS'23 [457])
- Development and application of MLLMs in modality alignment (arXiv'23 [1101])
- Instruction tuning for large language models: A survey (arXiv'23 [1631])
- Vision-language models for vision tasks: A survey (TPAMI'24 [1600])
- LLMs Meet Multimodal Generation and Editing (arXiv'24 [414])
- Visual Knowledge in Big Model: Retrospect and Prospect (arXiv'24 [1282])
- A Survey of Attacks on Large Vision-Language Models (arXiv'24 [756])
- A survey on hallucination in large vision-language models (arXiv'24 [763])
- The hallucination of MLLM: A Survey (arXiv'24 [35])
- Survey of forgetting in DL beyond continual learning (PAMI'24 [1317])
- A Comprehensive Survey on Evaluation of Multimodal LLMs (arXiv'24 [297])
- Visual Large Language Models for Generalized and Specialized Applications (arXiv'25 [695])
- Parameter-Efficient Fine-Tuning for Foundation Models (arXiv'25 [1587])

Learn from model beyond fine-tuning: A survey (NMI'25 [1694])
PEFT for Pre-Trained Vision Models: A Survey (arXiv'24 [1424])

2.2 Multimodal Pretraining

Multimodal Pretraining

SOHO (CVPR'21 [478]): Visual dictionary to transform image to semantics
PCME (CVPR'21 [175]): Modality samples as probabilistic in common space
ALIGN (ICML'21 [502]): Jointly learn from noisy image alt-text data
CLIP (ICML'21 [998]): Predict which caption goes with which image
ViLT (ICML'21 [580]): Lang and vision without conv or region supervision
Flamingo (NeurIPS'22 [7])
Fisher merge via maximizing joint likelihood of posteriors (NeurIPS'22 [866])
DenseCLIP (CVPR'22 [1004]): Pixel-text match and contextual information
[DeCLIP \(ICLR'22 \[698\]: Self, multi-view and nearest-neighbor supervision](#)
Learn robust audio representations from clip (ICASSP'22 [1364])
BLIP (ICML'22 [650]): Multimodal mixture of encoder-decoder with large-scale noisy image-text pairs
AudioCLIP (ICASSP'22 [376]): Audio-Head with text and image modalities
CLAP (ICASSP'23 [251]): Connects language and audio contrastive learning
Quantify the CLIP openness: extensibility and stability (ACL'23 [1010])
LLaVA ([761] NeurIPS'23): Large Language and Vision Assistant connects a vision encoder and LLM
CLIP-ViP (ICLR'23 [1463]): Omnisource cross-modal with video proxy
BLIP-2 (ICML'23 [649]): Pre-train a lightweight Querying Transformer
MiniGPT4 ([1732])
CHiLS (ICML'23 [916]): Hierarchical subclasses for CLIP via GPT
CLIPPO (CVPR'23 [1183]): Render the alt-text as image and contrast
SoftMask (CVPR'23 [946]): Softly masked visual feats + various samples
CLIP-REID ([673]): Learnable text tokens with two stage
APTM (ACM MM'23 [1502]): Attribute prompt and text matching
Visual resilience, uncertainty calibration, anomalous detect (NeurIPS'23 [1185])
CLAP (ICASSP'23 [251]): Natural language and audio via contrast
ImageBind (CVPR'23 [324]): Align six modality embedding
Meta-Transformer (arXiv'23 [1648]): A unified framework for multimodality
SPHINX (arXiv'23 [754]): Unfreezing LLMs, mixed weight, mixed task, and mixed visual embedding
Expanding Vocabulary brings indistinguishable text features (ACL'23 [1011])
Image is modeled as foreign language and unified mask (CVPR'23 [1274])
InstructBLIP (NeurIPS'23 [191]): Instruct-aware query transformer
LLaVA (NeurIPS'23 [761])
CogVLM (arXiv'23 [1280])
DCC (arXiv'24 [1275]): Divide, conquer and combine for high-resolution multimodality
PaLI (ICLR'23 [136]): A jointly-scaled MLLM

UNILM (NeurIPS'23 [458]): Aligning Perception with Language Models
 M2PT (arXiv'24 [1645]): An auxiliary transformer with for another modality
 Birdge (ICLR'24 [1316]): Knowledge graphs to connect unimodal models
 Honeybee ([83]): Flexible and locality-enhanced projector
 (CVPR'24 [760])
 VILA (CVPR'24 [741]) Large language instructed segmentation assistant
 Better training recipes and quality data for small LLM (arXiv'24 [1710])
 Language converges to representations isomorphic to vision (TACL'24 [1771])
 LISA (CVPR'24 [606])
 Local image augmentation, and Mixture of global Experts (arXiv'24 [1656])
[Leverage text to drive visual token align at language level \(arXiv'24 \[1114\]\)](#)
[LLaVA-HR \(arXiv'24 \[819\]\): Two visual pathways for images with different resolutions](#)
 (arXiv'24 [636])
 VILA² (arXiv'24 [275]): Self-augmenting and specialist-augmenting for VILA
 BLIVA (AAAI'24 [444]): Query embeddings and directly project patch
[LLaMA-VID \(ECCV'24 \[704\]\): Represent each frame with two distinct tokens: context token and content token](#)
 LLaVA- ϕ (arXiv'24 [1760])
 Concatenate visual tokens from a set of complementary vision encoders
 (arXiv'24 [1078])
 Cambrian-1 (arXiv'24 [1178]) A dynamic and spatially-aware connector that
 integrates high-resolution vision features
 LLaVA-UHD (ECCV'24 [373])
[Mix CLIP and vision-only SSL model features for LLaVA \(CVPR'24 \[1179\]\)](#)
 LaVIN (NeurIPS'24 [818]): Mixture-of-Modality Adaptation and routing
 LLaVA-Grounding (ECCV'25 [1591]): Add visual interaction + visual grounding modules
 VHM (AAAI'25 [934]): Versatile and Honest Vision Language Model for RS
 Janus (arXiv'24 [1351]) Decoupling visual encoding for unified multimodal
 understanding and generation

2.3 Multimodal Fushion

Fuse multiple modalities together.

FiLM (AAAI'18 [968]): Feature affine transformation via conditioning info
 Gated (AAAI'18 [566]): Gate modality via sigmoid non-linearity
 Modeling multimodal clues in a hybrid manner ([522])
 LMF (ACL'18 [799]): Low-rank fusion with modality-specific factors
 MFM (ICLR'19 [1182]): Modal discriminative and specific generative factors
 MISA (ACMMM'20 [403]): Reconstruct and fuse invariant and specific
 MBT (NeurIPS'21 [902]): Fusion bottleneck for multiple layers fusion
 Self-MM (AAAI'21 [1558]): Unimodal generation and weight adjustment
 Contrast multi-modal anchor tuples with optimizing aug (ICCV'21 [789])
 Modality gap by initialization and contrastive learning (NeurIPS'22 [730])

M2FNet (CVPR'22 [174]): A multi-head fusion and adaptive margin triplet
 GMC (ICML'22 [978]): Geometrically align complete and specific
 FDMER (MM'22 [1481]): Common and private encoders with discriminator
 Feat separation, brownian bridge, and geometric consistency (CVPR'23 [516])
 PMF (CVPR'23 [702]): Fuse uni-modal pretrained via disentangled prompt
 Feature and conversation in disentanglement and fusion (ACMMM'23 [631])
 Anchor graphs from different views not aligned (NeurIPS'22 [1267])
 DFERC ([632]): Dual-level disentangle for modality- and utterance- aspects
 i-Code (AAAI'23 [1515]): Pretrain modality encoder and multi-modal fusion
 Quantify multimodal interact: infor decomposition (NeurIPS'23 [725])
 DMD (CVPR'23 [707]): Decouple into modality-irrelevant/-exclusive spaces

2.3.1 Multimodal Conflict

Modalities have different optimization speeds, which obstacles the performance.

MAN (arXiv'20 [489]): Pre-train modality sub-net and attention decides
 Compute modality blending via overfitting behaviors (CVPR'20 [1281])
 UMT (arXiv'21 [232]): Modality failure via uni-modal teacher
 G-Blending (NeurIPS'21 [473]): Multimodality achieves smaller population
 Modality competition for the sub-optimality of joint training (ICML'22 [475])
 SHAPE (IJCAI'22 [441]): Margin contribution and cooperation degree
 OGMGE (CVPR'22 [966]): Gradient adaptively controls balance
[Greedy \(ICML'22 \[1370\]\): Modality dependence via conditional utilization](#)
 MSLR (ACL'22 [1523]):
 PMR (CVPR'23 [265]): Prototype stimulates slow modality
 MMCosine (ICASSP'23 [1449]): Modality-wise L_2 on feat and weights
 Balanced audiovisual data with modality discrepancies (arXiv'23 [1396])
 AGM (ICCV'23 [646]): Shapley mono-modal value for discrepancy ratio
 FAGM (ACMMM'23 [1304]): Para rebalance via dominance proportion
 MSRL (AAAI'23 [91]): Visual specific representations via reinforcement
 I²MCL (ACMMM'23 [1727]): Consider data difficulty and modality balance
 FULLER (ICCV'23 [476]): Calibrate multi-level gradient on task + modality
[Enhance sample-wise low-contributing modalities \(CVPR'24 \[1335\]\)](#)
 Alternately update modality and reconciliation regularization (ICML'24 [449])
 Alleviate modality preference and regulate essential component (ICLR'24 [1519])

2.3.2 Missing Modality

Handle the missing modality problem in multi-modal tasks

SMIL (AAAI'21 [843]): Modality reconstruction and feature regularization
 Transformer models are sensitive to missing modalities (CVPR'22 [842])
 Missing-aware prompt to alleviate missing modality (CVPR'23 [620])
 ShaSpec (CVPR'23 [1230]): Shared and specific learned features
 MMANet (CVPR'23 [1332]): Inter-class margin and combine weak modality

2.4 Hallucination & Attack & Unlearn

Backdoor and poisoning attacks significant threat the Multimodal learning

Training on unfiltered datasets intensifies poisoning attacks (ICLR'22 [77])

CleanCLIP (arXiv'23 [37]): CLIP learns spurious correlations

BiCro (CVPR'23 [1501]): Rectify binary noisy correspondence labels

Woodpecker (arXiv'23 [1541])

Objects that frequently appear in the visual instructions or co-occur with the image objects are obviously prone to be hallucinated by LVLMS (EMNLP'23 [691])

Learn robust text prompts via adversarial examples (CVPR'24 [657])

VCD (CVPR'24 [623]): [Contrastive distributions to calibrate output](#)

ICD (ACL'24 [1293]): Contrast from standard and instruction disturbance

VaCoDe (arXiv'24 [574]): Adaptively select aug with highest contrast

SWARM (CVPR'24 [1494]): Learns a trigger and switch prompt tokens

CroPA (ICLR'24 [820]): Update visual adv perturb with learnable prompt

M3ID (CVPR'24 [278]): [Amplify influence of ref image over language prior](#)

PATCH (arXiv'24 [1049]): [Insufficient extraction and visual decouple](#)

LLaVA-Align (arXiv'24 [1657]): Post-hoc debiasing and debias sampling

RITUAL (arXiv'24 [1346]): Original and transformed image in complement

MLLM becomes less reliant on visual information (NeurIPS'24 [1248])

VDGD (arXiv'24[323]): Grounding response generation in visual descr

HIO (arXiv'24 [90]): [Amplify contrast between hall and targeted tokens](#)

AvisC (arXiv'24 [1347]): [Identify blind tokens and contrast logits](#)

Subtracts an undesired dist. from original dist. (arXiv'24 [1750])

LVLMS suffer recency bias and authority bias (arXiv'24 [1404])

SeTAR (NeurIPS'24 [709]): Selective low-rank approx of weight matrices

[“Focus-on-Vision” prompt enhances visual data](#) (arXiv'24 [782])

ALCD (arXiv'24 [1436]): Redefine MIE as an identify-and-classify process

MARINE (arXiv'24 [1681]): Object grounding encoder enriches visual context

PAI (arXiv'24 [773]): Adjust and amplify attention weights to image tokens

[Overly detailed data leads to stop generation decrease](#) (arXiv'24 [1569])

Large Language Model Unlearning with control (arXiv'24 [1263])

CT²S (arXiv'24 [482]): [Preserve least important vision token to amplify hall](#)

Fine-Tuning LLM on new knowledge encourages hall (arXiv'24 [320])

PVP (EMNLP'24 [570]): [Counterfactual object, attribute, relation](#)

Enhance MLLM with detection and ocr (arXiv'24 [525])

Image weakened prompt and LLM-bias injection

ID(ICLR'25 [155]): Decode with self-consistency incorporated in objective

2.5 Visual Model Pretraining

Visual Model Pretraining

ViT (ICLR'21 [230]): Introduce transformer into classification

CPVT (CVPR'21 [171]): Replaces positional encoding with convolutions

CeiT (CVPR'21 [1564]): Add a depth-wise convolution in FFN
DeiT (ICML'21 [1180]): Distillation transfers inductive biases
ConViT (ICLR'21 [247]): Positional SA with convolutional inductive bias
LocalViT (arXiv'21 [711]): Incorporate 2D depth-wise convolutions
Swin (ICCV'21 [797]): Shifted window models global and boundary features
CvT (CVPR'21 [1362]): Utilize convolution to perform token embedding
CSWin (CVPR'22 [227]): Self-attention in horizontal and vertical stripes
MPViT (CVPR'22 [619]): Multi-scale Patches + Multi-Path structure
VARS (ICML'22 [1073]): Group into separate objects and select most salient
AbSViT (CVPR'23 [1071]): Top-down ViT variationally approximates AbS
LiVT (CVPR'23 [1462]): Masked generative pretraining and binary CE
Segment anything (arXiv'23 [584])
S² (arXiv'24 [1074]): Feature by large Multi-scale smaller pre-trained ones

2.6 Language Model Pretraining

Some Basic Papers.

Attention (NeurIPS'17 [1201])
Bert: Pre-training of deep bidirectional transformers (arXiv'18 [217])
Language models are unsupervised multitask learners (OPENAI'19 [999])
Language models are few-shot learners (NeurIPS'20 [55])
Scaling language modeling with pathways (arXiv'22 [168])
Palm 2 technical report (arXiv'23 [17])
GPT-4 Technical Report (arXiv'23 [922])
LLaMA: Open and efficient foundation language models (arXiv'23 [1181])
Vicuna (arXiv'23 [157])
Loss fails to explain and flatness correlates downstream (ICML'23 [762])
Data selection via importance resampling (NeurIPS'23 [1418])
LaCLIP (NeurIPS'23 [262]): Incorporate text aug via text rewriting
Progressively slides start state of reasoning from a demonstration end to its beginning (ICML'24 [1390])

2.7 Large Model FineTuning

Pre-trained Language Models Are Near Perfect for OOD (arXiv'23 [1194])
Enhance Multimodal capability for downstream tasks
Speciality vs generality: catastrophic forgetting in fine-tuning foundation models (arXiv'23 [750])
PEFT for large models: A comprehensive survey (arXiv'24 [388])
Adapter and connector do well in fine-tuning (ACL'24 [1726])
Parameter-Efficient Fine-Tuning for Foundation Models (arXiv'25 [1587])

2.7.1 Additive Fine-tuning

Adapters

Adapters (ICML'19 [433]): Serial adapter on self-attention and FFN layer
 CIAT ([1754])
 Krona ([248])
 LST (NeurIPS'22 [1535]): Intermediate activations via shortcut connections
 Tip-Adapter (ECCV'22 [1624]): Training-free CLIP-Adapter
 CLIP-Adapter (IJCV'23 [308]): Bottleneck layer and residual connections
 CoDA ([621]): sparse activation mechanism to improve inference efficiency
 Offsite-Tuning (arXiv'23 [1400]): Weight adapter and compressed emulator
[FDT \(CVPR'23 \[146\]: Learnable tokens for certain visual semantic concepts](#)
 CaFo (CVPR'23 [1626]): Utilize GPT-3, CLIP, DINO, DALL-E knowledge
 ViT with latent audio-visual hybrid adapter (CVPR'23 [752])
 SAN (CVPR'23 [873]): Side adapter for mask proposals and attention bias
 APE (ICCV'23 [1753]): Adaptive prior refinement from inter-class disparity
 MetaAdapter (NeurIPS'23 [149]): A lightweight residual to refine features
 GraphAdapter (NeurIPS'23 [684]): Dual graph contains structure information
 ViFi-CLIP (CVPR'23 [1006]): Fine tune or prompt CLIP to video domain
 VLAB (arXiv'23 [411]): Adapt and blend feature for video based on CLIP
 Cap4Video (CVPR'23 [1378]): Generated cap, intermediate interact, score
[FourierFT \(ICML'24 \[314\]: Treat M in spatial and learn spectral coeff.](#)
 CPR (CVPR'24 [101]): Conditional adapter and nearest neighbor rectification
 LLAMA-ADAPTER (ICLR'24 [1625]): Lightweight adaption prompt
 Adaptively ensemble layer-wise pre-trained and task-specific model (arXiv'24 [1019])
[HPT \(NeurIPS'24 \[1252\]\): Align specific proprioception and vision inputs to a short sequence of tokens](#)

Prompt

AutoPrompt (EMNLP'20 [1088]): Combine original inputs with trigger
 Compressing BERT by Progressive Module Replacing (EMNLP'20 [1434])
 LPAQA (ACL'20 [524]) Mining and paraphrasing-based to generate diverse prompts
 PrefixTuning (arXiv'21 [689]): Learnable vectors at keys and values
 LM-BFF (ACL'21 [309]): Automating prompt generation with fine-tuning
 Prompt Tuning (EMNLP'21 [626]): Store small task-specific prompt
 Continuous embedding space and manual initialization (NAACL'21 [1708]):
 LPT (arXiv'22 [780]): Leverage a prompt generator to obtain instance-aware prompt
 Xprompt ([839]): Eliminates negative prompt via hierarchical pruning
 SPT ([1751]): Learnable probabilistic gate whether to use the prompt propagated from previous or newly generated
 IPT (arXiv'22 [986]): Identify an intrinsic task subspace by auto-encoder

CoOP (IJCV'22 [1715]): Turns context words into a set of learnable vectors
VPT (ECCV'22 [507]): Learnable parameter in the input space
DoPrompt (arXiv'22 [1702]): Domain prompt learning and prompt adapter
DePT (arXiv'22 [311]): Hierarchical prompt alleviates error accumulation
Linear probing then full fine-tuning (ICLR'22 [601])
The visual prompt is effective and robust to distribution shift ([29])
CoCoOp (CVPR'22 [1714]): Prompt conditioned on query rather fixed on
UPT (arXiv'22 [1575]): Uniformly optimize prompts for different modalities
Linearly interpolate between zero-shot and fine-tuned (CVPR'22 [1349])
TPT (NeurIPS'22 [1091]): Marginal entropy with confidence selection
Temporal transformer and lightweight prompt vectors (ECCV'22 [534])
DualPrompt (ECCV'22 [1318]): Query task-invariant and specific
Design prompt via neural architecture search (arXiv'22 [1655])
DepthCLIP (ACMMM'22 [1629]): Zero-shot monocular depth estimation
DualCoOp (NeurIPS'22 [1125]): Encodes positive and negative prompts
background interpretation and context grading scheme (CVPR'22 [236])
InfoPrompt(arXiv'22 [1366]): Better prompt initialization and learn task-
relevance
SMoP (EMNLP'23 [165]): Sparsely activate short soft prompts via gate
Consider bias and diversity via zero-shot and fine-tuned (NeurIPS'23 [1730])
ViPT (CVPR'23 [1741]): Learns modal-relevant prompts to adapt
LPT (ICLR'23 [224]): Shared and group prompts for long-tailed
SP (CVPR'23 [130]): Semantic prompts adaptively tunes visual extraction
AdaMVMoE (ICCV'23 [123]): Automatically determine active experts
Pre-trained reveals sample difficulty via Mahalanobis (NeurIPS'23 [186])
(ICML'23 [591])
Learnable gates require prompt to focus on relevant blocks (ICML'23 [1545])
TOAST (arXiv'23 [1072]): Redirect the attention to task-relevant feature
PTP(EMNLP'23 [115]): Perturbation regularization to smooth landscape
Aprompt(EMNLP'23 [1264]): Learnable prompts to query, key, value
Automatically re-map the source label to the target (CVPR'23 [89])
DeFO (ICLR'23 [1224]): Decomposed feature prompt and additional layer
BIKE (CVPR'23 [1381]): Bidirectional cross-Modal knowledge exploration
Global models data distribution and local for discrimination (CVPR'23 [1321])
Pretrained language generates good semantic targets (AAAI'23 [1380])
Temporal model via arithmetic operations between frame (ICCV'23 [1379])
Task inference, identify and adaptive prediction (NeurIPS'23 [1255])
PromptStyler (ICCV'23 [159]): Prompt style diversity + content consistency
NPT (arXiv'23 [1734]): Language collapse and modality isomorphism ETF
ALIP (ICCV'23 [1490]): Language and description consistency gate
Combine prompt with an efficient adaptation network (arXiv'23 [1117])
Class visual prompt generated by text prompts and image token (TMM'23
[1426])
DualCoOP++ (TPAMI'23 [442]): Evidence guides dual context
Min transport cost and max prototypes and features mutual (ICML'23 [1155])
R-AMT (ICCV'23 [1696]): Identify key parameters and attach a binary mask

DAPT (ICCV'23 [158]): Text inter and Visual intra dispersion prompt
 TaI (CVPR'23 [371]): Double-grained prompt for coarse and fine grained
 Attention-based prompt-selection for CL (CVPR'23 [1097])
 MaPLe (CVPR'23 [561]): Couple prompts to ensure mutual synergy
[Test-Time prompting with distribution alignment \(NeurIPS'23 \[1031\]\)](#)
 Red circle directs the model attention to that region (CVPR'23 [1090])
 Black visual prompt via coordinator and perturbation (CVPR'23 [919])
[PromptSRC \(ICCV'23 \[562\]\): Mutual agreement with frozen and self-ensemble](#)
 KAPT (ICCV'23 [543]): Category-related external prompt via wikiPedia
 ProGrad (ICCV'23 [1729]): Updates prompt with aligned gradient to general
 E²VPT (ICCV'23 [151]): Learnable key-value prompts and visual prompts
 PLOT (ICLR'23 [100]): Learn multiple prompts via optimal transport
[KgCoOp \(CVPR'23 \[1521\]\): Learnable prompts closed to the fixed prompts](#)
 Replaces LLM descriptors with random character and word (ICCV'23 [1023])
 Prompt tuning uses pseudo labels generated by CLIP (NeurIPS'23 [874])
 Hierarchical Consistent and the Mean Treecut Accuracy (arXiv'23 [1377])
 Reformulate classification as generative masked visual token (arXiv'23 [735])
[Debias text embedding with a calibration project matrix \(arXiv'23 \[173\]\)](#)
 CMPA (ACL'23 [781]): Mutual exchange representation through multi-head
 RLFC (ICLR'23 [1687]): A novel reward function with CLIP for TTA
 Subspace prompt tuning and novel feature alignment (TCSVT'23 [837])
 VPT is good at similar data distribution and task objectives (ICLR'24 [379])
 PromptKD (CVPR'24 [713]): Pre-stored teacher text for distillation and
 inference
 ProText (CVPR'24 [560]): Learn text prompt from LLM
[Dept \(ICLR'24 \[1085\]\): Decompose prompt into a pair of low-rank matrices](#)
 (CVPR'24 [948])
[DePT \(CVPR'24 \[1608\]\): Capture base-specific knowledge in isolated space](#)
 Visual prompt and domain-specific prompts for text encoder (arXiv'24 [1253])
 Employ LLMs as constituent encoder for visual tokens (ICLR'24 [941])
 DAC (ICML'24 [1268]): CoOp has a larger textual distribution gap between
 the base and new classes
 BioCLIP (CVPR'24 [1108]): Fine-grained classifier for structured biology
 Cascade-CLIP (ICML'24 [697]): Align multi-level visual in cascaded way
 TAC (ICML'24 [693]): Leverage textual semantics of WordNet to facilitate
 image clustering
 Align between localized image areas and finer descriptions by LLM (ICML'24
 [651])
 TCP (CVPR'24 [389]): Transfers class textual embedding into class prompt
 ArGue (CVPR'24 [1167]): Attribute-Guided Prompt Tuning via LLM
 RESTORE (arXiv'24 [1514]): Explicit constraint on cross-modal consistency
 CoPrompt (ICLR'24 [1025]): Perturbed input with prompt + adapter
 CPT (ICML'24 [971]): Controllable prompt tuning for group fairness
 SaCo (CVPR'24 [1375]): Sample-wise affinity consistency
 ViP-LLaVA (CVPR'24 [62]): Linguistic interactions with visual markers
 DPL (CVPR'24 [69]): Domain-specific foundation with quaternion networks

DAMP (CVPR'24 [237]): Prompt to elicit domain-agnostic embeddings
 DPR (CVPR'24 [150]): Disentangle text prompt via LLM
 NegLabel (ICLR'24 [519]): Take negative labels from the corpus
 Temperature scaling improves calibration for VLM (ICML'24 [1184])
 An Empirical Study of CLIP Model Distillation (CVPR'24 [1473])
 Consistent Prompting for Rehearsal-Free Continual Learning (CVPR'24 [1673])
 TVP (CVPR'24 [1646]): Align feature and enrich task semantic
 Remove non-essential attention layers by entropy (CVPR'24 [744])
 STP (ICML'24 [1296]): Inferred token prototypes as the initial prompts
[Checkpoint recycling and hyperspherical adaptive MoE \(arXiv'24 \[1752\]\)](#)
 DOR (arXiv'24 arXiv'24 [1266]): Control unseen textual distribution divergence via zero-shot CLIP

2.7.2 Reparameterization

LoRA (ICLR'22 [438]): Injects trainable rank decomposition matrices
 AdaLoRA (arXiv'23 [1620]): SVD prunes unimportant update
 Factorize linear transformation as width- and depth-growth (ICLR'23 [1261])
 DoRA (ICML'24 [774]): Decompose weight into magnitude and direction
 LoRAMoE (ACL'24 [231])
 FLoRA (ICML'24 [393]): Constantly resamples the projection matrix
 LoRA+ (ICML'24 [402]): Simply set different learning rates for LoRA adapter matrices A and B
[MoSLoRA \(EMNLP'24 \[1376\]\): Learn the mixer with the original LoRA weights](#)
[SCMoE \(arXiv'24 \[1075\]\): Utilize unchosen experts in a self-contrast manner during inference](#)
[TeamLoRA \(arXiv'24 \[746\]\): Asymmetric structure and game-theoretic interaction](#)
 Visual and textual low-rank residual attention (arXiv'24 [1658])
 PEGO (ECCV'24 [440])
[Conducts rank-wise cluster by group MSUs from different LoRAs \(arXiv'24 \[1692\]\)](#)
 HydraLoRA (NeurIPS'24 [1165]): LoRA into A B and MoE B
 LoRA forgets less of the source domain (TMLR'24 [47])
 LoRA have new, high-ranking singular, intruder dimensions (arXiv'24 [1092])
 CorDA ([1507]): knowledge-preserved adaptation and instruction-previewed adaptation
 LoRA.rar (arXiv'24 [1069]): Merge LoRAs via Hypernetworks for Subject-Style
[LoRA learns learning response initiation and style tokens \(ICML'24 \[322\]\)](#)
[Full-parameter finetuning leads to knowledge degradation \(ICML'24 \[322\]\)](#)
 CLoRA [813]
 LoRAMoE [231]
 PRoLoRA (ACL'24[1269]): Partial Rotation Empowers More Parameter-Efficient LoRA

MixLoRA [1063]
 ResLoRA [1079]
 ACL’24 [1265]
 MeLoRA [1009]
 ShareLoRA (arXiv’24 [1104]): Shared Low Rank Adaptation
 LoRS (arXiv’25 [446]): Weight recompute and computational graph rearrangement for LoRA

2.7.3 Sparse Learning

CHILD ([1450])
 PST (IJCAI’22 [699]): Combine the data-free and data-driven criteria
 MoEfication (ACL’22 [1661]): Split FFNs into multiple experts with routers
 LT-SFT (ACL’22 [19]): Learn sparse, real-valued masks based on LTH
 Dual Lottery Ticket Hypothesis (ICLR’22 [33])
 ROSE (arXiv’22 [512]): First and Second-order selection for fine-tuning
[Energy and direction correlate with FT accuracy \(ICLR’23 \[1013\]\)](#)
 SAM (AAAI’23 [300]): Second-order approx with solvable optim
 LoSparse (ICML’23 [710]): Low-rank and structured sparse approximation
[Aggregate task-specific experts learned from similar tasks \(ICML’23 \[1352\]\)](#)
 LM-Cocktail: resilient tuning via model merging (arXiv’23 [1402])
 FISHDIP (EMNLP’23 [200]): Re-calculate Fish mask in each period
 PaFi (ACL’23 [732]): Train parameters with smallest absolute magnitude
 General instruction tuning helps alleviate the CF in LLMs (arXiv’23 [826])
[OFT \(NeurIPS’23 \[989\]\): Learn a layer-shared orthogonal transformation](#)
 SIFT (ICML’24 [1103]): Update partial components based on gradient norm
 Lingual-SMoE (ICLR’24 [1689]): Sparse moe with language guided routing
[TaSL \(ACL’24 \[286\]\) Task skill localization and consolidation](#)
[TaSLPLUS \(arXiv’24 \[285\]\): Identify task-relevant, shared specific param](#)
 MoFO ([143]): Update with largest momentum magnitudes in each block
[Flexible norm and relative angular adjust under soft ortho \(ICML’24 \[845\]\)](#)
 Layer adaptive sparsity ratio and layer-wise remove (CVPR’24 [738])
 GPS (CVPR’24 [1671]):
[SPU \(CVPR’24 \[1638\]\): Localize sparse updates to task relevant elements](#)
[CoFiTune \(ACL’24 \[1594\]\): Tree-search identifies gaining specialty](#)
 SAT (arXiv’24 [838]): Accelerate train by disregarding unimportant neuron
 AlphaEdit (ICLR’25 [271]): Null-Space Constrained Tuning anti-forgetting

2.7.4 Model Merge

Merge different models together
 Task-Arithmetic (ICLR’23 [485]): Combine task vectors through arithmetic
[TIES \(NeurIPS’23 \[1465\]\): Trim low-mag changes and resolve sign disagreements](#)
 Tailor (ICML’24 [1736]): Preserve pre-trained and selectively fine-tune
 Alter delta through loss diff via Riemann sums (arXiv’24 [1149])

[LoraHub \(COLM'24 \[451\]\)](#): Combine LoRA via Shiwa Evolver Searched
[DARE \(ICML'24 \[1555\]\)](#) Drops delta parameters and rescale remaining ones
[Twin-Merging \(NeurIPS'24 \[814\]\)](#) Share and exclude via SVD and MoE
[CART \(arXiv'24 \[164\]\)](#): Weight averaging via low-rank appro
[PCB \(NeurIPS'24 \[233\]\)](#): Intra+inter similarity for task importance
[EMR-Merging \(NeurIPS'24 \[452\]\)](#): [Elect, mask, rescale via task vector](#)
[SMILE \(arXiv'24 \[1146\]\)](#): Zero-shot sparse mixture of low-rank experts
[SUPERMERGE \(arXiv'24 \[1487\]\)](#): Hierarchical merging via learnable weight
 (arXiv'24 [856])
[ATM \(\[1717\]\)](#) Alternate tuning and merging
[LoRA Recycle distills a meta-LoRA \(arXiv'24 \[448\]\)](#)
[REMEDY \(ICLR'25 \[1735\]\)](#):

2.7.5 Data Sampler

Select helpful samples for the downstream task

LLMs are affected by SFT Data Composition (arXiv'23 [225])
 Gradient-based selecting high-quality and diverse data (ACL'24 [930])
[SDFT \(ACL'24 \[1518\]\)](#): [Fine-tune with a distilled dataset generated by itself](#)
[LESS \(ICML'24 \[1394\]\)](#):
[DLC \(arXiv'24 \[518\]\)](#): Learning complexity to quantify sample hardness
[ARF \(CVPR'24 \[380\]\)](#): [Regularize FT with a auxiliary textual contrast](#)
 Leverage LLMs to estimate data utility from small samples (arXiv'25 [415])

2.7.6 Efficient Inference

Speed up Large Model Inference

[LLaVA-PruMerge \(arXiv'24 \[1047\]\)](#): Class and spatial tokens similarity
[FastVLM \(arXiv'24 \[1621\]\)](#): Re-rank image tokens using [CLS] attention
[FastV \(ECCV'24 \[117\]\)](#):
[MaskLLM \(NeurIPS'24 \[270\]\)](#): Learnable prun crafts accurate N:M sparsity
[SparseseVLM \(arXiv'24 \[1647\]\)](#): Select visual-relevant text tokens and deter-
 mine sparsification ratio
[Victor \(arXiv'24 \[1341\]\)](#): Append learned visual register tokens
[VTW \(arXiv'24 \[753\]\)](#): Withdraw visual tokens at a certain layer
[GTP \(WACV'24 \[1457\]\)](#): [Sparse graph for tokens based on regeneration and](#)
[broadcasting difficulty](#)
[MustDrop \(arXiv'24 \[775\]\)](#): Measure token importance from vision encoding,
 prefilling, and decoding stage
[SF-LLaVA \(arXiv'24 \[1445\]\)](#): Capture spatial and temporal features
[VisionZip \(arXiv'24 \[1495\]\)](#): Dominant Token Selection and Contextual To-
 kens Merging
[p-MoD \(arXiv'24 \[1606\]\)](#): Mixture-of-Depths MLLM via Progressive Ratio
 Decay
[FrameFusion \(arXiv'24 \[299\]\)](#): Combine similarity-merge with importance-
 prune

[RLT \(NeurIPS’24 \[167\]\)](#): Compare temporally consecutive patches and removes redundant ones

2.7.7 Efficient Training

Efficient Training for Large Model

[GaLore \(ICML’24 \[1678\]\)](#) Memory-Efficient LLM Training by Gradient Low-Rank Projection

Weight normalize with low-rank and gradient low-rank ([arXiv’24 \[454\]](#))

2.7.8 Reinforcement Learning

[Bootstrapping Language Models with DPO Implicit Rewards \(arXiv’24 \[92\]\)](#)

2.8 Large Model Purning

([NeurIPS’89 \[612\]](#))

([ICLR’17 \[888\]](#))

Reduce storage and computation by magnitude order ([383])

Prune filters with relatively low weight magnitudes ([ICLR’17 \[645\]](#))

Deep learning for computer vision: A brief review ([CIN’18 \[1211\]](#))

[LTH \(ICLR’19 \[295\]\)](#): Networks contain subnetworks (winning tickets)

Pruning can be useful as an architecture search paradigm ([ICLR’19 \[800\]](#))

[PRAC \(ICML’21 \[1660\]\)](#): Data and model sparsity co-design

[Dynamic sparsity is channel-level sparsity learner \(NeurIPS’23 \[1539\]\)](#)

[DiffPurning \(ACL’20 \[359\]\)](#): Learnable binary mask to model weights

Most occupations exhibit some degree of exposure to GPTs ([arXiv’23 \[252\]](#))

Essential sparsity with a sharp dropping point ([NeurIPS’23 \[493\]](#))

[SparseGPT \(ICML’23 \[296\]\)](#): Estimate the Hessian matrix, adapting the traditional OBS

[BESA \(ICLR’24 \[1447\]\)](#): Minimize block-wise error and optimize sparsity rates across layers

[Wanda \(ICLR’24 \[1121\]\)](#): Purn by weight magnitude and feature activation

Capture spatial semantics and long-range temporal for video LLaVA ([arXiv’24 \[1445\]](#))

[AFT \(arXiv’25 \[708\]\)](#) Synthesize multiple draft responses into a single answer

2.9 Language Model Safety

LLMs via natural prompts can enable more straightforward attack tactics

LLMs memorize and leak individual training examples ([USENIX’21 \[78\]](#))

An attacker can indirectly perform Prompt Injection attacks ([arXiv’23 \[337\]](#))

Larger models morally self-correct—to avoid harmful outputs ([arXiv’23 \[304\]](#))

Auto evaluation: fluency, correctness, citation quality ([EMNLP’23 \[310\]](#))

Detect jailbreak prompts by safety-critical gradients ([ACL’24 \[1419\]](#))

RigorLLM ([1567]): Energy-based augmentation and fuse KNN
Instructive decoding: Instruction-tuned large language models are self-refiner from noisy instructions ([578])
Weak-to-strong backdoor attack contrastive KD (arXiv’24 [1684])
Supervised FT on uncertainty-augmented predictions (arXiv’24 [88])
DAM (arXiv’24 [1489]): Dual mask optimization to identify a shared and safety-aware subspace
(arXiv’25 [1301])

2.10 Large Model Distillation

Large Model with Knowledge Distillation

f -DISTILL (ACL’23 [1342]): A family of sequence-level distilling
MiniLLM (ICLR’24 [353]): Reverse KLD is more suitable for KD on generative language models
DISTLLM (ICML’24[585]): Skew KD and adaptive off-policy approach
ATKD (ACL’24 [1705]): More attention to diverse learning of hard-to-learn tokens
PromptKD (arXiv’24 [567]): Distill student-friendly knowledge for generative language Models via prompt tuning
MiniPLM (ICLR’25 [354]): Leverage the differences between large and small LMs to enhance the difficulty and diversity of the training data

2.11 Phenomenon

Attention sink and so on.

[Activations are input agnostic and extremely rare quantity \(COLM’24 \[1120\]\)](#)
Super weights induce rare and large activation outliers (arXiv’24 [1556])
[Attention sink like key biases, extra attention scores \(NeurIPS’24 \[352\]\)](#)
(NeurIPS’24 [1330])
(arXiv’24[1331])
Lower layers transfer general visual features into question tokens
Middle layers transfer specific visuals to question token
Higher layers propagate to prediction position (arXiv’24 [1670])
SEG token mainly contributes to semantic correspondences (arXiv’24 [984])
SFT memorizes training and struggles to generalize OOD (arXiv’25 [169])
RL improves underlying visual recognition capabilities (arXiv’25 [169])

Chapter 3

Finance

3.1 Behavior Finance

Some intriguing papers!

- Foreign investors is momentum but domestic is converse (JFE'00 [340])
- Disposition effect: tendency to sell winners and hold losers (JF'01 [342])
- Distance, language, culture influence stock holdings and trades (JF'01 [341])
- Investors with overconfidence and sensation seeking trade more (JF'09 [343])
- Status concerns lead households to track neighbors' wealth (RFS'14 [422])

3.1.1 Kahneman and Tversky

A friendship that changed our minds. Chinese Link

- Regard a sample randomly drawn as highly representative (Psychol. Bull.'71[1189])
- Representativeness judges via essential property (Cogn. Psychol.'72 [536])
- Person evaluates classes frequency by availability (Cogn. Psychol.'73 [1190])
- Prediction appears most representative of evidence (Psychol. Rev.'73 [537])
- Representativeness, availability and anchoring adjustment (Science'74 [1191])
- Uncertain values prediction and confidence intervals assessment ('77 [538])
- People asymmetrically assess loss and gain (Econometrica'79 [539])

3.1.2 Richard Thaler

"I will try and spend it as irrationally as possible."

- Mental accounting organizes, evaluates, tracks finance (MS'85 [1163])

3.2 Fintech

Some papers about Fintech

- Common risk factors in the returns on stocks and bonds (JFE'93 [258])
- Dissecting anomalies (JF'08 [259])

Optimal versus naive diversification (RFS'09 [204])
Predictability of returns and cash flows (ARFE'11 [586])
Presidential address: Discount rates (JF'11 [176])
International stock return predictability (JF'13 [1005])
An SVM-based approach for stock market trend prediction (IJCNN'13 [749])
The superview of return predictive signals (RAW'13 [335])
A five-factor asset pricing model (JFE'15 [260])
... and the cross-section of expected returns (RSF'16 [398])
Tree-based conditional portfolio sorts (SSRN'16 [890])
Information aggregation about cross-section of stock returns (RFS'17 [737])
Characteristics with independent information on month returns (RFS'17 [336])
Firm characteristics and Chinese stocks (JMSE'18 [508])
Characteristics are covariances: Unification of risk and return (JFE'19 [557])
Machine Learning-driven Quantamental Investing (CIE'19 [634])
Shrinking the cross-section (JFE'20 [595])
Cost-sensitive selection via deep reinforcement learning (TKDE'20 [1654])
Federated meta-learning for fraudulent credit card detection (IJCAI'21 [1701])
Identify foundations, themes, research from bibliometric (JBEF'21 [329])
Model stock as internal and relational attributes (FinNLP'22 [671])
Machine learning in the Chinese stock market (JFE'22 [622])
Postfundamentals price drift in capital markets (MS'22 [25])
A probabilistic dynamic factor model based on variational autoencoder (AAAI'22 [242])
Doubly robust via prediction of treatment and outcome (AISTATS'23 [474])
An Online and Adaptive Factor Model Based on Hierarchical and Regime-Switch VAE (IJCAI'23 [1340])
Financial news announcements and a multi-factor model (WWW'24 [685])

3.2.1 Dacheng Xiu

Link.

Empirical asset pricing via machine learning (RFS'20 [348])
(Re-)Imag(in)ing Price Trends (CBRP'20 [509])
Autoencoder asset pricing models (JE'21 [349])

3.3 Interesting

Some papers that I like.

The Matthew effect in science (Science'68 [875])
The rich get richer and the poor get poorer [1058]

3.4 Game Theory

Some papers about Game-Theory

A value for n-person games ([1053])

Game theory: analysis of conflict ([899])
The kernel of a cooperative game: ([201])
Stackelberg solution for nonzero-sum two-player (JOTA'73 [1093])
Cooperative games on combinatorial structures ([49])

Chapter 4

Contrastive Learning

We show the report linear evaluation accuracy on ImageNet for ResNet50

4.1 Theory

Theoretical analysis of self-supervised learning especially contrastive learning

- Similar pairs are assumed to be the same latent class (ICML'19 [1039])
- Good views retain task-relevant rather irrelevant info (NeurIPS'20 [1174])
- Alignment of positive; uniformity of feats on hypersphere (ICML'20 [1271])
- Multi embed, each of which is invariant to all but one aug (ICLR'21 [1403])
- CL is hardness-aware; τ controls the strength of penalties (CVPR'21 [1225])
- Reconstruction SSL with appro conditional independence (NeurIPS'21 [615])
- No single self-supervised method dominates overall (CVPR'21 [253])
- Positive align; Centers diverge; Augmentation concentrate (arXiv'21 [470])
- CL shares low/mid but overfit semantic info of data (ICLR'21 [1683])
- Aug removes features associated with spurious dense noise (ICML'21 [1344])
- Aug creates chaos for intra-class and CL climbs ladder (ICLR'22 [1312])
- Dimensional collapse: vectors spans a lower-dim subspace (ICLR'22 [530])
- Different SSL methods have a similar gradient structure (CVPR'22 [1156])
- Two hardness-aware independently with dual temperature (CVPR'22 [1584])
- Local geometry decides the learning behavior of SSL (arXiv'22 [1770])
- A probabilistic model selects optimal negative samples scale (IJCAI'22 [1354])
- Incorporate function class bias into transfer bounds (ICML'22 [1038])
- AggNCE (NeurIPS'22 [183]): Aligns the mean feature of multiple views
- CL probably over-fits to shared information between views (CVPR'22 [1231])
- Treat each aug differently and learn hierarchical invariance (CVPR'22 [1605])
- MLP Improves transferability of unsupervised pretraining (CVPR'22 [1309])
- Analyze through distributionally robust optimization (NeurIPS'23 [1365])
- Asymmetric create a consistent rank difference (ICLR'23 [1768])
- Optimizing the robust contrastive losses with automatic temperature individualization (ICML'23 [990])

[Stronger inflation should be accompanied by weaker aug \(ICLR'24 \[1311\]\)](#)

4.1.1 Yuandong Tian

Link.

The intermediate layer update is governed by covariance (arXiv'20 [1175])
The nonlinear learning dynamics of non-contrastive SSL (ICML'21 [1168])
Dimensional collapse: vectors spans a lower-dim subspace (ICLR'22 [530])
Siamese keeps a lower variance in target than source (CVPR'22 [1287])

4.1.2 Tengyu Ma

Link.

Expansion magnifies local to global class consistency (ICLR'21 [1322])
Spectral decomposition on augmentation graph (NeurIPS'21 [395])
SSL are more robust to imbalance than supervised learning (ICLR'22 [758])
CL disentangles domain and class rather learn domain-invar (ICML'22 [1059])
Pos has more connection with same class in cross-domain (NeurIPS'22 [396])
Dimension and sample contrast are two sides of same coin (ICLR'23 [315])
Standard InfoNCE is equivalent to similar graph spectral clustering (ICLR'24 [1145])

4.2 Baseline

Some famous methods

InstDisc (CVPR'18 [1388]): Instance Discriminate and memory bank
DeepCluster (ECCV'18 [80]): Cluster features and assign pseudo-labels
CPC (arXiv'18 [921]): Conduct predictive learning on sequential information
InvaSpread (CVPR'19 [1531, 1529]): End-to-end learning with siamese net
MOCO (CVPR'20 [408]): Query should be similar to key than others - 60.6
SimCLR (ICML'20 [125]): Data aug and nonlinear transformation - 69.3
PIRL (CVPR'20 [885]): Semantic representations should be invariant - 63.6
CMC (ECCV'20 [1170]): Mutual infor maximization for different views - 66.2
BYOL (NeurIPS'20 [339]): Online and target networks interact - 74.3
SwAV (NeurIPS'20 [81]): Clusters data and enforce consistency - 71.8
SimSiam (CVPR'21 [133]): Predictor+stop-gradient is important - 71.3
W-MSE (ICML'21 [254]): Align positive and whitening to avoid collapse
TWIST (arXiv'21 [1223]): Prediction consistency + sharpness + diversity
E-SSL (ICLR'22 [199]): Highlight the invariance and equivariance in SSL
MEC (NeurIPS'22 [783]): SSL need admit the maximum entropy
UniVCL ([1150]): View the predictor from the graph perspective
GSG (NeurIPS'23 [613]): Implicitly leverage asymmetric relation
Learn periodic information with customized periodicity (ICLR'23 [1508])

4.2.1 Transformer

Self-supervised learning in transformer

DINO (ICCV'21 [82]): SSL provides new properties to Vision Transformer

TokenCut (CVPR'22 [1307]): Token graph for bi-partition based on DINO

CuTLER (CVPR'23 [1288]): Discover multiple object masks with MaskCut

4.3 Dimension Decorrelation

Redundancy-reduction principle

Barlow Twins (ICML'21 [1576]): Cross-correlation reduces redundancy

VICReg (ICLR'22 [39]): Invariance + Variance + Covariance

TiCo (arXiv'22 [1742]): Min transform invariance and covariance contrast

ARB (CVPR'22 [1633]): Group + Shuffle Dimension Alignment

ZeroCL (ICLR'22 [1636]): Instance and Dimension Alignment

MetaMask (NeurIPS'22 [652]): Dimension redundancy and confounder - 73.9

VICRegL (NeurIPS'22 [40]): VICReg with local visual features

4.4 Positiveness

Discover positiveness in SSL

MixCo (NeurIPS'20 [576]): Semi-positives from pos and neg mixup

MSF (CVPR'21 [592]): Leverage kmeans to enlarge positive pairs - 72.4

NNCLR (ICCV'21 [246]): Positive from nearest neighbor - 74.5

i-Mix (ICLR'21 [617]): Virtual labels for a batch and mixture

ContrastiveCrop (CVPR'22 [965]): Semantic-aware and center-suppressed

Un-Mix (AAAI'22 [1066]): Soft distance concept on label space in CL

SDMP (CVPR'22 [1012]): Mixed images as additional positive pairs

RINCE (AAAI'22 [421]): Exploit information about a similarity ranking

SNCLR (ICLR'23 [316]): Highly-correlated instances as soft neighbors - 75.3

4.5 Negativeness

Discover negativeness in SSL

DCL (NeurIPS'20 [172]): Correct the sampling bias of negative examples

MoChi (NeurIPS'20 [541]): Mix hardest negative with itself and query

ReSSL (NeurIPS'21 [1700]): Instance relational consistency with different

aug

NonSemNeg (NeurIPS'21 [317]): Negative samples only preserve non-semantics

HCL (ICLR'21 [1016]): Hard negative are mapped nearby but should be far.

RELIC (ICLR'21 [886]): Enforce invariant prediction of proxy targets - 70.3

CO2 (ICLR'21 [1323]): Similarities between query and negatives - 68.0

ISD (ICCV'21 [1161]): Use a soft similarity for negative images - 69.8

DecoupCL (arXiv'21 [1533]): Decouple positive and negative samples - 69.5

4.6 Positiveness and Negativeness

Leverage positive and negative simultaneously

FeatTrans (ICCV'21 [1749]): Harder positives and diversified negatives

MetAug (ICML'22 [653]): Margin-injected meta feature augmentation

4.7 Cluster and Prototypes

Cluster and prototypes for SSL

DnC (ICCV'21 [1169]): Train individual experts to recover local consistency

PCL (ICLR'21 [654]): Introduce prototypes to encode semantic structures

GCC (CVPR'21 [1704]): Clustering-friendly features and compact cluster

HCSC (CVPR'22 [370]): Capture hierarchical semantic structures

4.8 Scene Images

Complicated scene images for SSL

ORL (NeurIPS'22 [1415]): Pretrained SSL discovers object-level semantic

UniVIP (CVPR'22 [717]): Scene-scene, scene-instance, and instance-instance

4.9 Detection and Segmentation

Prior SSL focuses on classification, but ignores detection and segmentation

InsLoc (CVPR'21 [1477]): Foreground is copy-pasted onto background.

PixPro (CVPR'21 [1421]): A pixel-to-propagation consistency task

DetCo (CVPR'21 [1413]): Multi-level supervision + Global and local CL.

PLRC (CVPR'22 [32]): Point-level region contrast for localization and holism

4.10 Distillation

SSL faces knowledge distillation

Compress (NeurIPS'20 [1]): Mimic relative similarity between the samples

SimDis (arXiv'21 [346]): A simple distillation baselines

SEED (ICLR'21 [276]): Mimic similarity distribution inferred by teacher

ProtoCPC (ICLR'22 [616]): Prototypical probabilistic discrepancy for CL

BINGO (ICLR'22 [1438]): Bags related instances by matching similarities

DisCO (ECCV'22 [312]): A consistency regularization to align embedding

4.11 Supervised Learning

Introduce CL into supervised learning

SCL (NeurIPS'20 [565]): Supervised contrastive learning

HiMulConE (CVPR'22 [1635]): Leverage all labels and preserve hierarchy

LOOK (ICLR'22 [287]): Consider intra-class difference via kmeans
THANOS (ICML'22 [118]): Balanced spread + class permutation invariance

4.11.1 Long Tailed

Handle long tailed problem

Labels bring supervision and bias in class-imbalanced (NeurIPS'20 [1511])
KCL (ICLR'21 [544]): k-positive contrastive learning
SDCLR (ICML'21 [523]): A self-competitor online by pruning target
Hybrid-SC (CVPR'21 [1260]): Hybrid network includes SCL and CE
PaCo (ICCV'21 [185]): Parametric class-wise learnable centers to re-balance
TSC (CVPR'22 [675]): Force all classes to maintain uniformity
BCL (CVPR'22 [1744]): Class-averaging and class-complement
SSL is more robust to dataset imbalance (ICLR'22 [758])

Chapter 5

Graph Learning

5.1 Preliminary

Base research about graph learning

- Spectral networks and locally connected networks on graphs (ICLR'14 [56])
- DeepWalk (SIGKDD'14 [969]): Structure by truncated random walks
- node2vec (SIGKDD'16 [344]): Learn continuous representations for nodes
- Fast localized convolutional filters on graphs (NeurIPS'16 [203])
- GCN (ICLR'17 [583]): Localized first-order approximation of spectral gc
- GraphSAGE (NeurIPS'17 [377]): Generate embeddings from neighborhood
- GAT (ICLR'18 [1202]): Specify different weights to different neighborhoods
- GIN (ICLR'19 [625]): GNN under the neighborhood aggregation framework
- DeepGCNs (ICCV'19 [641]): Residual/dense connect and dilated conv
- EdgeConv (TOG'19 [1308]): Capture local geometric and recover topology
- PNA (NeurIPS'20 [180]): Multiple aggregators and degree-scalers
- OrthoGConv (AAAI'22 [362]) Maintain orthogonality of feature transform

5.2 Analysis

Some analysis of graph learning

- GCN model is a special form of Laplacian smoothing (AAAI'18 [667])
- SGC (ICML'19 [1359]): Remove nonlinearity and collapsing weight matrices
- DGN (NeurIPS'20 [1713]): Normalize group of similar nodes
- MADGAP (AAAI'20 [95]): Low information-to-noise ratio leads over-smoothing
- CSSL (AAAI'21 [1578]): Contrastive learning pretrain alleviates overfitting
- Training GNNs for long-range face over-squashing (ICLR'21 [10])
- Graph-MLP (arXiv'21 [445]): Contrast neighbor via adjacency information
- AIR (SIGKDD'22 [1640]): Divide GNNs into transform and propagate
- A dynamic graph attention variant for GAT (ICLR'22 [54])
- DeCorr (SIGKDD'22 [526]): Feature overcorrelation with over-smoothing
- OrthoReg (arXiv'22 [18]): The orthogonality of node correlation matrix

Deep GCNs suffer from poor gradient flow during training (NeurIPS’22 [494])
 MLPInit (ICLR’23 [385]): Accelerate GNN via converged PeerMLP initialize
 PMLP (ICLR’23 [1475]): MLP in training and GNN in testing
 GLWI (SIGKDD [788]): Long-range propagation and handling stray nodes
 CONTRANORM (ICLR’23 [366]): Optimization-induced normalization layer

5.3 Augmentation

Graph learning with data augmentation

DropEdge (ICLR’20 [1022] TPAMI’22 [461]): Randomly removing edges
 GraphMix (AAAI’21 [1207]): Train a FCN jointly with the GNN
 GAUG (AAAI’21 [1688]): Edge predictors encode class-homophilic structure
 KDGA (NeurIPS’22 [1367]): Leverage KD to reduce negative augmentation
 FLAG (CPVR’22 [590]): Aug node features with gradient-based adversarial
 GAME (NeurIPS’22 [767]): Larger version difference of high than low freq

5.4 Self-Supervised Learning

Graph learning with self-supervised learning

DGI (ICLR’19 [1203]): Local mutual max on graph’s patch representations
 InfoGraph (ICLR’20 [1112]): Maximize mutual info from overall and local
 GRACE (ICML’20 [1758]): Remove edge, mask node: structure, attribute
 GCC (SIGKDD’20 [988]): Graph contrastive coding for GNN pre-training
 GraphCL (NeurIPS’20 [1549]): Four types of graph augmentations for GCL
 MVGRL (ICML’20 [399]): Node and graph (first-order and graph diffusion)
 InfoGCL ([1435]): Information-aware graph contrastive learning
 GCA (WWW’21 [1759]): Identify important edges and feature dimensions
 AD-GCL (NeurIPS’21 [1131]): Adversarial graph augmentation for GCL
 CCA-SSG (NeurIPS’21 [1593]): Canonical correlation analysis with GCL
 GraphLoG (ICML’21 [1446]): Hierarchical prototypes for global-semantic
 JOAO (ICML’21 [1548]): Automatically select data augmentation for GraphCL
 BGRL (ICLRW’21 [1162]): Graph contrastive learning with BYOL
 MGSSL (NeurIPS’21 [1663]): Motif captures subgraphs (functional group)
 AFGRL (AAAI’22 [618]): Positive by local structure and global semantics
 SAIL (AAAI’22 [1554]): Intra- and inter-graph knowledge self-distillation
 HGRL (CIKM’22 [112]): Preserve original feat and capture general neighbor
 GraphMAE (SIGKDD’22 [431]): Recover feature with scaled cosine error
 BYOV (WSDM’22 [1550]): InfoMin and InfoBN guides prior learning
 GSCL (arXiv’22 [912]): Rank node representation in the neighborhood
 ProGCL (ICML’22 [1393]): Most negatives with larger similarities are false
 SimGRACE (WWW’22 [1392]): GNN model with its perturbed version
 BAGE (TPAMI’22 [1630]):
 PGCL (TNNLS’22 [743]): Model semantic structure of graph via clustering

5.5 Federated Graph Learning

Graph learning with federated scenario

FedGraphNN (ICLRW'21 [406]): Federated Learning Benchmark for GNN

GCFL (NeurIPS'21 [1414]): Graph clustered FL based on gradient

FedSage (NeurIPS'21 [1610]): Subgraph FL with missing neighbor generation

A federated GNN for privacy-preserving personalization (NC'22 [1355])

FedStar (AAAI'23 [1142]): Sharing degree-based structure embedding

FGSSL (IJCAI'23 [463]): Federated graph semantic and structural learning

5.6 Long-tailed Learning

5.7 Confidence Calibration

Calibration of GNNs

Improve GNN calibration to increase trustworthiness (ICMLW'19 [1160])

CaGCN (NeurIPS'21 [1292]): A topology-aware post-hoc GNNs calibration

GATS (NeurIPS'22 [436]): Nodewise temperature scaling with attention

DR-GST (WWW'22 [759]): High-confidence nodes bring distribution shift

5.8 Compression

5.8.1 Pruning

GNN with pruning for training and inference problem

UGS (ICML'21 [127]): Co-simplify adjacency and weight with Lottery Tickets

5.8.2 Knowledge Distillation

Knowledge distillation in graph learning

LSP (CVPR'20 [1509]): Local structure preserving for GCN Distillation

TinyGNN (SIGKDD'20 [1466]): Peer-aware module with neighbor distillation

RDD (SIGMOD'20 [1639]): Define node and edge reliability in a graph

G-CRD (arXiv'21 [533]): Preserve global topology via contrastive learning

GFKD (IJCAI'21 [213]): GNN Distill without observable graph

GLNN (ICLR'22 [1632]): Eliminate graph dependency via distilled MLPs

LLP (arXiv'22 [372]): Distill relational knowledge for MLP

BGNN (arXiv'22 [374]): Complementary knowledge from different GNNs

MSKD (AAAI'22 [1583]): Learn from multiple GNNs with attention weight

GKD (NeurIPS'22 [1476]): Neural Heat Kernel to encapsulate geometric

5.8.3 Self-Knowledge Distillation

Self-knowledge distillation in graph learning

IGSD (arXiv'20 [1592]): Iterative self-distillation to contrast graph pairs

GNN-SD (IJCAI'21 [137]): Shallow layers as supervision for deeper layers

GSDN (arXiv'22 [1368]): Graph self-distillation on neighborhood

5.9 Multimodal Learning

Multimodal learning with Graph learning

MMGCN (ACMMM'19 [1338]): Multi-modal GCN for Recommendation

5.10 HyperGraph

HyperGraph learning

HyperGCL (NeurIPS'22[1334]): Fabricate and generate contrastive views for

5.11 Application

Some applications on graph learning

PinSage (KDD'18 [1543]): GCNN for web-scale recommender systems

GAMENet (AAAI'19 [1045]): Medical recommend via drug-drug interact

SemGCN (CVPR'19 [1682]) Local, global semantic relations for Pose

Graph convolutional neural network in social networks (NaNA'19 [1172])

5.12 Survey

Some survey about Graph Learning

Survey of graph: Problems, techniques, and applications (TKDE'18 [60])

A Comprehensive Survey on Graph Neural Networks (TNNLS'20 [1386])

Survey on Graph Structure Learning (arXiv'21 [1757])

Graph self-supervised learning: A survey (TKDE'22 [792])

Data augmentation for deep graph learning: A survey (arXiv'22 [220])

Chapter 6

Knowledge Distillation

6.1 Logits

Logits-level knowledge distillation

- KD (arXiv'15 [417]): Transfer knowledge from big to small model
- ESKD (ICCV'19 [160]): More accurate teacher is not a good teacher
- TAKD (AAAI'20 [882]): An assistant to bridge the gap
- PAD (ECCV'20 [1650]): Adaptive sample weighting: easy is highlighted
- Tf-KD (CVPR'20 [1565]): KD is a learned label smoothing regularization
- GLD (ICCV'21 [581]): global and local logits + relational information
- MT-BERT (ACL'21 [1353]): Multi-teacher for language model compression.
- SFTN (NeurIPS'21 [945]): Learn the teacher which is friendly to students
- DKD (CVPR'22 [1675]): Target distillation and non-target distillation
- DIST (NeurIPS'22 [460]): Correlation-based loss for intra and inter-class
- ATS (NeurIPS'22 [687]): Guidance, Smooth and class discriminability
- NKD (arXiv'22 [1516]): Rethinking knowledge distillation via cross-entropy
- MLLD (CVPR'23 [529]): Prediction + input and category correlation
- DOT (ICCV'23 [1674]): A larger momentum for distillation acceleration
- NormKD (arXiv'23 [156]): Add normalization before softmax
- Normalized KD via $S_t + V_t - \text{mean}(S_t)$ (ICCV'23 [1520])
- LogitStand (CVPR'24 [1124]): Weighted Z-score before softmax
- On-the-Fly Sample Selection to filter out rank-distorted samples (CVPR'24 [481])

6.2 Feature

Feature-level knowledge distillation

- FitNet (ICLR'15 [1021]): Map the hidden layer from student to teacher
- AT (ICLR'17 [587]): Optimize FitNet and use the attention map of features
- Overhaul (ICCV'19 [416]): Feature transform with margin ReLU
- SSKD (ECCV'20 [1437]): Combine self-supervision with KD

FNKD (ECCV'20 [1444]): Feature normalized replace unified temperature
 HSAKD (AAAI'21 [1472]): Capture hierarchical self-supervised knowledge
 SemCKD (AAAI'21 [97]): Assign each tea layers for stu via attention
 DKMF (TPAMI'21 [1229]): Mimick both feature magnitude and direction
 ReviewKD (CVPR'21 [120]): Cross-stage connection
 SCKD (ICCV'21 [1756]): Relative gradient direction control distillation
 CID (NeurIPS'21 [212]): Capture sample and class and remove bias
 SimKD (CVPR'22 [96]): Align feature and reuse teacher classifier
 MGD (ECCV'22 [1517]): Mask student feature and generate teacher feature
 KF (ECCV'22 [1506]): Decompose pretrained network into factor networks
 Improved feature distillation via projector ensemble (NeurIPS'22 [144])
 Neural Collapse Inspired Knowledge Distillation (AAAI'25 [1634])
 FOFA (arXiv'25 [742]): Region aware attention and prompt tuning blocks

6.3 Relation

Relation-level knowledge distillation

GiftKD (CVPR'17 [1536]): FSP computes inner product between layers
 RKD (CVPR'19 [953]): Pairwise and ternary relations of examples
 IRG (CVPR'19 [787]): Instance relationships and feature transformation
 CCKD (ICCV'19 [957]): Transfer correlation between instances
 SP (ICCV'19 [1187]): Preserve the pairwise similarities
 CRD (ICLR'20 [1171]): A contrastive objective to transfer knowledge
 Intra-category and inter-category structure distillation (ECCV'20 [148])
 LONDON (ICCV'21 [1048]): Minimize the distance via Lipschitz constants
 HKD (CVPR'21 [1718]): Distill holistic knowledge based on graph
 CRCd (CVPR'21 [1743]): Inter-sample relation and representation learning
 REFILLED (TPAMI'22 [1525]): Comparison match and adaptive local kd

6.4 Self-Distillation

Self Distillation Paper

BAN (ICML'18 [302]): An ensemble of multiple students generations
 DDSD (AAAI'19 [1451]): Align feature and logits on data distortion
 Snapshot (CVPR'19 [1479]): Signal from prior iteration rather generation
 MultiExit (ICCV'19 [973]): Encourages early exits to mimic later exits
 BYOT (ICCV'19 [1616]): Deeper knowledge is squeezed into shallow ones
 SAD (ICCV'19 [430]): Attention maps from deeper as targets for lower layers
 CS-KD (CVPR'20 [1571]): Align output of different sample of same class
 Learnable attentive transfer to identify informative channels (ECCV'20 [1422])
 Self-Distillation as Instance-Specific Label Smoothing (NeurIPS'20 [1668])
 PS-KD (ICCV'21 [572]): Previous epoch model as teacher to conduct KD
 BAKE (arXiv'21 [318]): Inter-sample affinities weight to provide soft label
 KR (AAAI'21 [221]): Alleviate optim conflict and maintain class relations

FRSKD (CVPR'21 [501]): Utilize an auxiliary self-teacher network
AFA (NN'21 [1423]): Identify relevant channels and spatial feature
DLB (CVPR'22 [1062]): Constrain half of batch coinciding with previous
Zipf (ECCV'22 [724]): Use prediction to get soft label under Zipf distribution
MixSKD (ECCV'22 [1474]): MixUp alignment on feature and logits
Tf-FD (ECCV'22 [655]): Reusing channel and layer-wise features
FER (AAAI'22 [1399]): Restrict behavior on the correctly-classified samples
Utilize shallow branches and promote different branches (IF'23 [802])

6.5 Theory

Some explanation about knowledge distillation.

Compress the function learned by a complex into a smaller (KDD'06 [57])
LUPi (JMLR'15 [1200]): Similarity control and knowledge transfer
RNN (ICASSP'16 [1153]): Simple is effectively to guide complex model
Unifying distillation and privileged information (ICLR'16 [803])
Convolution models need to be deep and convolutional (ICLR'17 [1195])
The teacher needs to be more tolerant (lower accuracy) (AAAI'19 [1478])
Data geometry, optimization bias and strong monotonicity (ICML'19 [972])
In noise-free data, teacher label smoothing hurts student (NeurIPS'19 [897])
Rescale student gradient and inject prior knowledge (arXiv'20 [1147])
KD brings various visual concepts and stable optimization (CVPR'20 [154])
In noisy data, teacher label smoothing benefits student (ICML'20 [816])
label smooth decrease variance and lower mean prediction (ICLR'21 [1067])
MSE does not elongate representation compared with KL (IJCAI'21 [579])
Good student doesn't imply good distillation fidelity (NeurIPS'21 [1107])
Rethink soft label: sample-wise bias-variance tradeoff (ICLR'21 [1712])
More and different knowledge points + stable optimization (TPAMI'22 [1622])
Link KD to information bottleneck principle (NeurIPS'22 [1218])
Pruned teacher is more effective than the original teacher (ECCV'22 [949])
SHAKE (NeurIPS'22 [658]): Extra shadow head as a proxy teacher
PEBKD (ECCV'22 [214]): Student adaptively find blind knowledge region
Teacher knowledge of some classes is undistillable (NeurIPS'22 [1755])
Good DA reduces covariance of tea-stu cross-entropy (NeurIPS'22 [1239])
Enhanced augment and larger datasets boost vanilla KD (arXiv'23 [394])
Training average does not work but KD works (arXiv'23 [9])

6.6 Survey

Some surveys about knowledge distillation.

Knowledge Distillation: A Survey (IJCV'21 [331])
Knowledge distillation: A review and new outlooks (PAMI'21 [1257])

Chapter 7

Robustness

7.1 Parameter Vulnerability

Parameter vulnerability is related to robustness and generalization.

- Gradient corruption via maximizing first-order estimation (AAAI'21 [1126])

- Finetune adversarial model on non-robust- critical module (ICCV'23 [1745])

7.2 Adversarial Attack

Adversarial attacks evaluate the robustness of models before deployment.

- BadNets (arXiv'17 [350]): Identify vulnerabilities in ML supply chain

- Modifying the discrete graph structure (ICML'18 [190])

- MI-FGSM (CVPR'18 [228]): Momentum iteration to an ensemble of models

- Adversarial examples improve image recognition (CVPR'20 [1412])

- TI (CVPR'19 [229]): Translation-invariance for transferable adv samples

- Small hypocritical perturb hinder adversarial training (NeurIPS'22 [1157])

7.3 Adversarial Defense

Defense against adversarial attack.

- HGD (CVPR'18 [734]): Error amplification and regularize top layers

- Adversarial examples are not bugs, they are features (NeurIPS'19 [486])

- MMC (ICLR'20 [937]): Induce dense feature via Max-Mahalanobis center

- Ensemble several pretraining tasks, which boosts robustness (CVPR'20 [126])

- Adversarial robustness is sensitive to some basic settings (ICLR'21 [939])

- PatchGuard (USENIX'21 [1398]): Robustly detects and masks corruption

- Robust and acc could be reconcilable by properly defined (ICML'21 [936])

- Grafting (ICML'22 [128]): Linearize insignificant ReLU for certifiable robust

- RR (CVPR'22 [940]): ϵ -error R-Con and $\frac{1}{2-\epsilon}$ Con rejector for separability

- IAT (CVPR'23 [226]): Align adversarial samples with those in high-likelihood

Generalist (CVPR'23 [1242]): Bi-expert for natural and robust generalization

AdapPatch (ICLR'23 [983]): Backdoor latent separability assumption fails

7.3.1 Adversarial Robustness Distillation

Training small robust models via knowledge distillation.

RSLAD (ICCV'21 [1769]): Soft labels on natural and adversarial examples

7.4 Backdoor Attack

Inject certain data patterns without altering behavior on the clean data.

BadNets (arXiv'17 [350]): Identify vulnerabilities in the ML supply chain

Patterned static and targeted adaptive perturbation mask (arXiv'17 [134])

Backdoor embedding in CNN via invisible perturbation (arXiv'18 [733])

Badnets: Evaluating backdooring attacks on DNN Access'19 [351]
(arXiv'19 [1188])

CompBack (SIGSAC'20 [740]): Composite by mixing benign features

Triggers exhibit artifacts in the high-frequency spectrum (ICCV'21 [1580])

ISSBA (ICCV'21 [696]): Sample-specific trigger via encoder-decoder

TA-LBF (ICLR'21 [30]): Flip critical bits for backdoor and not hurt others
(arXiv'21 [76])

(SP'22 [503])

Backdoor learning: A survey (TNNLS'22 [694])

Anchoring (ICLR'22 [1666]): Logits consistency on clean data.

BadPrompt (NeurIPS'22 [65]): Sample-wise adaptive trigger optim

FIBA (CVPR'22 [288]): Amplitude trigger in the frequency domain

Gradient matching, data selection, and re-training (NeurIPS'22 [1106])

Marksman (NeurIPS'22 [223]): Class-conditional generative trigger

Verify embedded external features, style transfer (AAAI'22 [712])

Handcraft parameters in filters to maximize activation separation ([426])

FTROJAN ([1273]): Frequency trigger with small pixel-wise perturbations

A uniform color space shift for all pixels as trigger (CVPR'23 [517])

BadCLIP ([729]): Dual-embedding guided backdoor for CLIP

7.5 Backdoor Defense

Distinguish or recover benign DNNs from backdoored ones.

Classify under Gaussian is certifiably robust to adversary (ICML'19 [177])

ADP (ICML'19 [938]): Ensemble diversity via Entropy and DPP

[Control gradient norm and orientation in poison and clean](#) (arXiv'20 [428])

ANP (NeurIPS'21 [1358]): Backdoor is sensitive to adversarial perturbation

CAS (ICLR'21 [34]): Suppress channel-wise activation for redundancy

CIFS (ICML'21 [1469]): Suppress NR and align PR channels

ABL (NeurIPS'21 [700]): Backdoor are faster to learn than clean examples

DPA (ICLR'21 [627]): Partition training set with multiple base classifiers
Trigger features are sparsely encoded in a few channels (NeurIPS'22 [63])
Backdoor forms a hyperplane via piece-wise linear (NeurIPS'22 [1313])
Trap via auxiliary reconstruction and replace classifier (NeurIPS'22 [1233])
Select poisoned samples via transformations sensitivity ([131])
ARGD (ECCV'22 [1391]): Reflect att feats correlations of different orders ([777]):
BagFlip (NeurIPS'22 [1644]): Model-agnostic certified via randomized smooth
DBD (ICLR'22 [455]): Decouple via backbone SSL and classifier fine-tuning
UBW (NeurIPS'22 [690]): Abnormal model behaviors are not deterministic
DEFEAT (CVPR'22 [1691]): Adaptive perturb + representation restrict ([355])
RNP (ICML'23 [701]): Neuron-level unlearning and filter-level recovering
PBE (CVPR'23 [893]): Progressively purify via untargeted adversary attack
CBD (CVPR'23 [1664]): Minimize mutual info with confounding represent
Clean distribution-aware regularizer via fisher (arXiv'23 [553])
[FT-SAM \(ICCV'23 \[1747\]\): Finetune backdoor-related norms by sharpness](#)
Actively deviate the classifier weights from origin (NeurIPS'23 [880])
Insert a learnable neural polarizer to filter trigger (NeurIPS'23 [1748])
Without train structure, backdoor is indistinguishable from nature (ICML'23 [559])
Contrast representations shifts before and after randomized channel shuffling of clean and trigger features ([64])
Decouple benign and enlarge differences with backdoor features (ICLR'24 [1453])

Chapter 8

Long-Tailed Learning

8.1 Class Re-balancing

Balance the training sample numbers of different classes during the training

Focal Loss (ICCV'17 [747]): Explore class prediction hardness for re-weighting

CB (CVPR'19 [188]): Effective number to approximate the expected scale

Bal Meta SoftMax (NeurIPS'20 [1008]): Label ratio and sample via validation

BAGS (CVPR'20 [705]): Group softmax on classes with similar scale

LADE (CVPR'21 [429]): Label distribution disentangling loss

Seesaw (CVPR'21 [1250]): Mitigation factor and compensation factor

BEV (CVPR'23 [1297]): Prediction with category-wise variation

8.2 Information Augmentation

Introduce additional information into model training

RIDE ([1291]): Multiple experts and expert assignment module

CKT (CVPR'22 [944]): Identify similar classes to compose rare ones

Contrastive learning and multivariate Gaussian distribution (ICCV'23 [1619])

Feature fusion from head to tail (AAAI'24 [663])

8.3 Module Improvement

Explored methods to improve network modules in long-tailed learning

Decoupling (ICLR'20 [545]): Square-root and progressively-balanced

MARC (ACML'22 [1310]):

Feature uncertainty representation with geometric information (IJCV'24 [850])

Chapter 9

Time Series Forecasting

9.1 Normal Task

Autoformer(NeurIPS'21 [1363]): Auto-Correlation discovers sub-series similarity and aggregates similar subseries

PatchTST (ICLR'23 [911]): Time series segmentation and channel-independence

SAMformer (ICML'24 [484]): Sharpness-aware minimization and channel-wise attention

iTransformer (ICLR'24 [790]): Embed series independently to variate token

Pathformer (ICLR'24 [121]): Integrate both temporal resolution and temporal distance

9.2 Fintech Task

Chapter 10

Colourful

10.1 Data Augmentation

Some papers about Mixup

Mixup (ICLR'18 [1589]): Convex combinations of pairs examples and labels
NLPmixup (arXiv'19 [360]): WordMixup + senMixup
ManiMix (ICML'19 [1206]): Semantic interpolations brings boundary
CutMix (ICCV'19 [1570]): Patches are cut/pasted and labels are also mixed
Remix (ECCV WorkShop'20 [166]): Disentangle mix factor of image label
AttentiveCutMix (arXiv'20 [1213]): Replace discriminative via attention
PuzzleMix (ICML'20 [571]): Saliency and local statistics for mixup
PatchUp (arXiv'20 [277]): Mask out contiguous blocks of feature map
Fair Mixup (ICLR'21 [892]): Interpolate samples between the groups
Label smooth improves generalization and attack perform (ICML'21 [556])
SaliencyMix (ICLR'21 [1192]): Saliency guided data aug
MixMo (ICCV'21 [1003]): Mix multiple inputs with patches
Mixup correlates to adversarial loss and reduces overfitting (ICLR'21 [1611])
Alignmix (CVPR'22 [1204]): Geometrically align two images in feature space
Transmix (CVPR'22 [113]): Not all pixels are created equal
TeachAug (CVPR'22 [1132]): Informative transformed images via a teacher
TokenMix (ECCV'22 [764]): Token masking and label with activation map
OpenMix (CVPR'23 [1738]): Mix ID and outlier with reject class
Mixup is more beneficial in learning rare features (arXiv'23 [1772])

10.2 Interesting

Some papers are interesting (The Review of Contemporary Fiction'14 [418])
Evaluation of adaptive mixtures of competing experts (NeurIPS'90 [917])
Mixture of experts: (Artificial Intelligence Review'14 [865])
Generative adversarial nets (NeurIPS'14 [330])
Joanna Ruocco. Dan ([418])

Projected Gradient Descent (arXiv'17 [854])
 SGD acts as an implicit regularizer (ICLR'17 [1582])
 Arbitrary style transfer with adaptive instance norm (CVPR'17 [471])
 MaxEnt (NeuIPS'18 [244]): Maximum-Entropy in fine-grained classification
 Enhance generalization capacities via ibn-net (ECCV'18 [931])
 Model extracts distribution-specific and -agnostic signals (ICLR'19 [1232])
 Unforgettable examples seem to play little performance (ICLR'19 [1177])
 OctConv (ICCV'19 [138]): Low- and high-frequency features separately
 Loss scales as power-law with model, dataset, compute size (arXiv'20 [552])
 DSBN (ECCV'20 [1043]):
 Rethink K-NN (arXiv'21 [505]): k-NN as an indication of examples difficulty
 MIMO (ICLR'21 [400]): Ensemble the predictions made by the subnetworks
 Distribution smoothing acknowledges effects of nearby targets (ICML'21 [1513])
 CHILD-TUNING (EMNLP'21 [1450]): Mask the gradients during backward
 CAN (EMNLP'21 [506]): Prior distribution fixes prediction in low confidence
 Free Lunch for Few-Shot Learning: Distribution Calibration (ICLR'21 [1497])
 Data valuation via NTK without optimizing network (ICML'22 [1387])
 Task Centroid Projection Removing features dimension along the task centroid (NeurIPS'22 [1442])
 MAETTT (NeurIPS'22 [303]): MAE for one-sample learning
 BatchFormer (CVPR'22 [432]): Learn sample relationships from mini-batch
 Learn distinct environment by auxiliary information (NeurIPS'22 [751])
 HCR (CVPR'22 [1139]): Similar structures on feature and logits hyperspheres
 NoisyTune (ACL'22 [1356]): Add limited noise before finetuning
 Multilingual language decouples into sensitive and neutral (EMNLP'22 [85])
 Equivariance and invariance inductive bias for limited data (ECCV'22 [1272])
 Self-supervised learning of visual graph matching (ECCV'22 [755])
 Vision GNN: An Image is Worth Graph of Nodes (NeurIPS'22 [381])
 GTA-Seg (NeurIPS'22 [528]): Disentangle pseudo on backbone and predictor
 CoCs (ICLR'23 [847]): View the image as points and employ clustering
 Non-Parametric Networks for 3D Point Cloud Analysis (CVPR'23 [1627])
 Reducing training data by examining generalization influence (ICLR'23 [1500])
 UAV-OD (CVPR'23 [1251]): UAV object detection via freq disentanglement
 Data Rights, Data Pricing, and Privacy Computing (Engineering'23 [1440])
 Subpocket proto and global interact with biochemical prior (ICML'23 [1662])
 Inadequately better fully Pre-training in transfer (CVPR'23 [205])
 PRG (ICCV'23 [243]): History class distribution and transitions
 PED (ICCV'23 [681]): Dynamic representations to decline potential energy
 Adjust correlation via pre-process and unfairness mitigation (ICML'23 [1018])
 Extreme feature redundancy of pretrained vision for few-shot (arXiv'23 [825])
[Meta distribution energy for autoeval \(ICLR'24 \[961\]\)](#)
 Proper noise in pre-training can benefit in-domain tasks and hurt out-of-domain tasks (ICLR'24[103])
 Generate task-specific class-discriminative frequency mask (ICCV'23 [152])
 LabelWave (ICLR'24 [1566]): Train prediction changes identify early stop

(CVPR'24 [4])

10.3 Tricks

Tricks for Deep learning

- Batch Norm (ICML'15 [487]): Reduce internal covariate shift
- CAM (CVPR'16 [1711]): Highlight the class-specific discriminative regions
- Layer Norm (arXiv'16 [26]): Remove dependencies between samples
- Instance Norm (arXiv'16 [1193]): Remove instance-specific contrast infor
- Weight Norm (NeurIPS'16 [1030]):
- Self-Ensembling (ICLR'17 [607]): Exponential moving average of predictions
- Mean Teachers (NeurIPS'17 [1158]): Averages weights rather predictions
- Cosine Norm (ICANN'18 [817]): Reduce neurons variance
- Group Norm (ECCV'18 [1383]): Divides the channels into groups
- Dive into deep learning: Tools for engagement (19 [997])
- BN introduces regularization and information leakage (arXiv'21 [1384])

10.3.1 Attention

Attention module on different aspects

- SENet (CVPR'18 [439]): Explicitly model channels interdependencies
- NL (CVPR'18 [1289]): Weighted sum of the features at all positions
- BAM (BMVC'18 [950]): Infer attention map along channel and spatial
- [CBAM \(ECCV'18 \[1348\]\): Infer attention map along channel and spatial](#)
- DANet (CVPR'19 [298]): Position and Channel attention
- CCNet (ICCV'19 [477]): Criss-cross attention

10.3.2 Label Smoothing

Some papers about label smoothing

- Label Smoothing (CVPR'16 [1134]): Smooth hard target distributions
- OLS (TIP'21 [1585]): Soft labels based on model prediction
- Negative label smoothing for label noise (ICML'22 [1328])
- ELS (ICLR'23 [1653]): Domain discriminator outputs soft probability

10.4 Domain Adaptation and Generalization

10.4.1 Domain Adaptation

Some papers about domain adaptation

- Analysis of representations for domain adaptation (NeurIPS'07 [43])
- Learning bounds for domain adaptation (NeurIPS'07 [51])
- Domain adaptation with multiple sources (NeurIPS'08 [860])
- A theory of learning from different domains (ML'10 [42])
- Algorithms and theory for multiple-source adaptation (NeurIPS'18 [420])

DA translation with domain invariant and specific info (NAACL'19 [347])
DADA (ICML'19 [963]): Disentangle domain-specific feats from class
SHOT (ICML'20 [723]): Freeze classifier via information max
DRMD-Net (CVPR'20 [1228]): Disentangled rep and independent feature
IDM (ICCV'21 [195]): Generate intermediate domain via source and target
G-SFDA (ICCV'21 [1499]): Structure cluster and sparse domain attention
Encourage label consistency with high local affinity (NeurIPS'21 [1498])
SAN-SAW (CVPR'22 [958]): Semantic alignment and distributed alignment
DIFEX (TMLR'22 [808]): Phase as invariance and cross-domain correlation
GRDA (ICLR'22 [1460]): Graph characterizes domain relations
DIPE (CVPR'22 [1222]): Domain-balance for invariance and specification
SFDA (NeurIPS'22 [1496]): Contrast for feature cluster and assignment
CDAN (TNNLS'23 [844]): Regularize domain difference via copula distance
CLUDA (ICLR'23 [926]): Domain-invariant, contextual rep for time series
Classifier uniformity and transferability and discriminability of feature (ICLR'24 [1488])

10.4.2 Domain Generalization

Some papers about domain generalization

Combining multiple datasets has positive performance (ECCV'12 [564])
Domain generalization via invariant feature representation (ICML'13 [895])
Domain-adversarial training of neural networks (JMLR'16 [305])
DG via conditional invariant representations (AAAI'18 [692])
CROSSGRAD (ICLR'18 [1050]): Directly perturb inputs by loss gradient
Domain-specific classifiers and domain prediction (ICIP'18 [857])
Learning De-biased Representations with Biased Representations ([28])
Meta-learning algorithm for domain generalisation (AAAI'18 [639])
IRM (arXiv19[22]): Minimize invariant risk via entire invariant predictor
G2DM (arXiv'19 [8]): One-vs-all domain discriminators for divergences
JiGen (CVPR'19 [79]): Self-supervised signals to solve a jigsaw puzzle
Random and evolution-based augmentation search (ICCV'19 [1210])
REx (arXiv'20 [597]): Extrapolated domains and training risks variance
CSD (ICML'20 [977]): Common and specific component via low rank
Deep-SLR (TMI'20 [979]): Multi domain-specific with structured low-rank
Domain generalization via Entropy Regularization (NeurIPS'20 [1685])
DMG([87]): Masking disentangles domain-invariant and -specific info
UniReP (CVPR'20 [1084]): Confidence-aware and feature decorrelation
DIRT(NeurIPS'21 [907]): StarGAN transforms data density among domains
mDSDI (NeurIPS'21 [59]): Meta learn domain-specific
Concat domain information as supplementary information (CVPR'21 [245])
RandConv (ICLR'21 [1461]): Random conv creates new domains samples
MixStyle (ICLR'21 [1716]): Mix feature statistics to synthesize novel
GLT (CVPR'21 [1695]): Multiple prototypes with different granularity
RaMoE (CVPR'21 [194]): Voting mixture dynamically with decorrelation
IDGM (arXiv'21 [1082]): Maximize gradient inner for alignment

PCL (CVPR'22 [1522]): Proxy-based contrastive learning
 DARLING (CVPR'22 [1643]): UDG ignores domain-related features
 DNA (ICML'22 [170]): Classifier ensemble with pruned Jensen-Shannon
 DSU (ICLR'22 [680]): Model domain uncertainty with synthesized feature
 Fishr (ICML'22 [1001]): Matches the gradient variances across domains
 Independent train and average based bias-variance-covar (NeurIPS'22 [1002])
 A survey on domain generalization (TKDE'22 [1246])
 TRO (ICLR'23 [985]): Integrate distributional topology for OOD resilience.
 Domain prototype contrast and domain expert classifier (arXiv'23 [1586])
 MAD (CVPR'23 [992]): Two-branch classifier: biased and general
 DCG (CVPR'23 [829]): Cast DG as a convex game between domains
 DRM (SIGKDD'23 [1651]): Domain-specific head with online model steering
[DomainDrop \(ICCV'23 \[361\]\): Suppress domain-sensitive channels](#)
 GSNR (ICCV'23 [878]): Discard high gradient-signal-to-noise ratio elements
[Generalizable Mixture-of-Experts \(ICLR'23 \[635\]\)](#)
 PALM (ICLR'24 [804]): Mixture prototypes and reciprocal neighbor soft weights

10.5 Test Time Adaptation and Training

10.5.1 Test Time Adaptation

Some papers about test time adaptation

T3A (NeurIPS'21 [491]): Pseudo-prototype and reduces prediction entropy
 TENT (ICLR'21 [1220]): Minimize entropy via updating normalization layers
 SADE (NeurIPS'22 [1649]): Aggregate expert by maxing prediction stability
 MEMO (NeurIPS'22 [1617]): Minimize marginal entropy for predictions
 SADE (NeurIPS'22 [1649]): Skill-diverse expert and test-time aggregation
 EATA (ICML'22 [913]): Active sample selection and Fisher regularizer
 Conjugate pseudo-label with meta-learned losses (NeurIPS'22 [332])
 VMP (NeurIPS'22 [531]): Perturb weight with variational Bayesian
 SPN (IWSCFF'22 [952]): Entropy minimization of the predicted foreground
 NHL (CVPR'23 [1152]): Unsupervised Hebbian with learned neuro-modulator
 ITTA (CVPR'23 [116]): Learnable consistency loss for test-time adaptation
 SAR (ICLR'23 [914]): Sharpness-aware and reliable entropy minimization
 TeCo (ICLR'23 [1534]): Min global entropy and Regulate local coherence
 Hyper-parameter, model selection, and batch dependency (ICLRW'23 [1677])
 Alignment and uniformity for TTA feature revision (CVPR'23 [1270])
 TIPI (CVPR'23 [905]): Be invariant under predefined transformations
 AdaNPC (ICML'23 [1652]): Recall k closed samples from memory
 TAST (ICLR'23 [496]): Ensemble scheme with nearest neighbor information
 A survey on test-time adaptation under distribution shifts (arXiv'23 [722])
 AdaMerging (ICLR'24 [1482]): Entropy minimization for merging coefficient

10.5.2 Test Time Training

Some papers about test time training

TTT (ICML'20 [1129]): Turn a single test into a self-supervised

10.6 Mixture-of-View-Experts

Mixture-of-View-Experts

Adaptive mixtures of local experts (NeurCompt'91 [492])

Hierarchical mixtures of experts and the EM (NeurCompt'94 [532])

Mixture of experts in multiclass classification (NeurNet'99 [114])

Deep mixture of experts via shallow embedding (UAI'20 [1295])

Hash layers for large sparse model (NeurIPS'21 [1020])

Base layers: Simplifying training of large, sparse models (ICML'21 [629])

Scaling vision with sparse mixture of experts (NeurIPS'21 [1014])

Tricks for training sparse translation models (arXiv'21 [238])

MoE ViT for multi-task with model-accelerator codesign (NeurIPS'22 [268])

Efficient scaling of language models with mixture-of-experts (ICML'22 [235])

Sparse MoE as Dropout Scaling Dense Self-Slimmable (ICLR'23 [129])

Autonomy-of-Experts Models via L2-Norm (arXiv'25 [828])

10.7 Overconfidence and Conformal Prediction

Over-parameterized deep neural networks face over-confidence and Conformal Prediction.

Concrete problems in AI safety (arXiv'16 [14])

Predictive uncertainty estimation using deep ensembles (NeurIPS'17 [608])

Evaluate predictive uncertainty under dataset shift (NeurIPS'17 [923])

Label Smoothing (CVPR'16 [1134]): Smooth hard target distributions

Confidence Penalty (ICLR Workshop'17 [967]) Penalizing low entropy

Remedy the miscalibration by temperature scaling (ICML'17 [358])

Deep ensembles and adversarial training for robustness (NeurIPS'17 [609])

CE encourages the output scores dramatically distinctive (CVPR'19 [410])

DNNs trained with mixup are better calibrated (NeurIPS'19 [1164])

Uncertainty on the IID setting and distributional shift (NeurIPS'19 [924])

- Accuracy and quality of uncertainty degrades with increasing dataset shift.

- Calibration and accuracy on IID test fail under dataset shift.

Dirichlet Calibration (NeurIPS'19 [599]): Multiclass calibration method

GER (ACL'20 [869]): A parameterized family of regularizers

Flooding (ICML'20 [488]): Make the loss float around the flood level

Focal loss reduces NLL overfit and improves calibration (NeurIPS'20 [896])

Hard improve calibrated result via post-hoc calibration (NeurIPS'21 [1221])

MoCaD (NeurIPS'21 [1428]): Calibrate bias-only model for ensemble

DecisionCalibration (NeurIPS'21 [1686]): Predict is indistinguishable to truth

[A \$K + 1\$ -way softmax based on energy functions \(ICCV'21 \[1302\]\)](#)

CPC (CVPR'22 [153]): $C(C-1)/2$ pairwise binary classification
Flat minima is beneficial for failure prediction (ECCV'22 [1737])
LogitClip ([1327]): Clamp logit norm for noisy over-confidence
DDNN (ICCV'23 [1733]): Distilled data leads poor calibration
Overfit more heavily on low proximity sample (NeurIPS'23 [1427])
Optimize the temperature value toward an optimal set that satisfies the marginal coverage
Attention sink like key biases could be non-informative (NeurIPS'24 [352])

10.8 Optimization Loss Space

Some papers about gradient analysis

([290])
([12])
([13])
([864])
[954]
Small batch to flat minima due to inherent gradient noise (ICLR'17 [558])
High gradients similarity cause performance degradation (AISTATS'18 [1538])
Similarity reflects perception and not priori norm (NeurIPS'19 [86])
Stiffness captures resistance to deformation by gradient (ICML'19 [292])
High gradient confusion may lead slow convergence (ICML'20 [1032])
Large initial learning rate steers to better loss surface (ICLR'20 [497])
Widen task local minima avoids catastrophic forgetting (NeurIPS'20 [883])

10.9 Neural Collapse

Features are linearly separable, collapsed to C-d simplex ETF (PNAS'20 [942])
Centers and classifiers will be simplex equiangular tight frame in balance.
Classifier randomly initialized as ETF and training fixed (NeurIPS'22 [1510])
Central path, where the linear classifier stays MSE-optimal (ICLR'22 [384])
FSCIL (ICLR'23 [1512]): Pre-assign an optimal feature-classifier alignment
CeCo (CVPR'23 [1707]): Context correlation and imbalanced dist.
FEDETF (ICCV'23 [716]): A simplex ETF for all clients
Compact within-class and distinct between-class (AISTATS'23 [785])

10.10 Sharpness-Aware Minimization

Simultaneously minimizing loss value and loss sharpness

SAM (ICLR'21 [291]): Seek parameters in neighbor with uniformly low loss
GSAM (ICLR'22 [1764]): Surrogate gap as equivalent sharpness measure
SSAM (NeurIPS'22 [876]): Sparse perturbation by binary mask to accelerate
FSAM (EMNLP'22 [1706]): Fisher mask to obtain the sparse perturbation
SAF (NeurIPS'22 [234]): Take the update trajectory in the past epoch

GA-SAM (EMNLP'22 [1665]): Gradient for different para and period
LookSAM (CVPR'22 [795]): Compute SAM periodically and reuse direction
FedASAM (ECCV'22 [66]): SAM and stochastic weight averaging
SAM maximization brings from the better objective (ICML'22 [16])
SWGM (CVPR'23 [1262]): Align gradient between empirical and perturbed
SAM reduces feature rank at different layers (NeurIPS'23 [15])
DISAM (ICLR' 24[1623]): Perturbation focus more on domains with higher convergence degree

10.11 Recommendation

Basic methods for the recommendation.

NCF (WWW'17 [413]): Model latent features of users and items
LightGCN (SIGIR'20 [412]): Neighbor aggregation for collaborative filtering
Adap- τ (WWW'23 [111]): Normalize embedding with adaptive temperature

10.12 Generative Model

Gaussian distribution is the most naive generative model in machine learning

Train Invertible Neural Network by Info Bootleneck (NeurIPS'20 [20])
SDGM (ICLR'21 [401]): Train GMM by sparse Bayesian learning
Generative classifiers for trustworthy classification (CVPR'21 [852])

10.13 Masked Image Modeling

Masked image modeling different fields.

MAE (CVPR'22 [407]): Randomly mask large parts and reconstruct value.
CAE (arXiv'22 [132]): Separate the encoder and decoder in MAE
SdAE (ECCV'22 [140]): A multi-fold masking strategy with self distillation
I-JEPA (arXiv'23 [24]): Predict representations of various blocks in query

10.14 Medical AI

When Deep learning faces the Medical field.

HotProtein (ICLR'23 [124]): Protein structure-aware and factorized sparse

10.15 Survey

Some surveys

A Survey on Transfer Learning (TKDE'09 [928])
Recent advances and trends in visual tracking: A review (NC'11 [1486])
Deep learning for person re-identification (TPAMI'21 [1528])
Deep learning for generic object detection: A survey (ICCV'20 [765])

Image segmentation using deep learning: A survey (TPAMI'21 [881])
Salient object detection in the deep learning era (TPAMI'21 [1277])
A survey on deep learning for video segmentation (TPAMI'22 [1284])

10.16 Mang Ye

Link.

Aug invariant and ins spreading for softmax embedding (PAMI'20 [1529])
DL for person re-identification: A survey and outlook (PAMI'21 [1528])
Collaborative refining for REID with label noise (TIP'21 [1527])
Structure-aware position transformer for visible-infrared REID (TIP'22 [94])
Federated cross-correlation and instance similarity (PAMI'23 [466])
Disentangled prototype with transformer for sketch-photo (PAMI'23 [93])
Channel Augmentation for Visible-Infrared Re-Identification (PAMI'23 [1530])

10.17 Wenguan Wang

Link.

GPNN (ECCV'18 [982]): Learn human-object interactions by graph parsing
PAGE-Net (CVPR'19 [1283]): Pyramid attention and salient edges
COSNet (CVPR'19 [809]): Frame correlation and global co-attention
Disentangle motion blur between foreground and background (ICCV'19 [1068])
AGNN (ICCV'19 [1279]): Attentive GNN for video graphs confusion
MuG (CVPR'20 [811]): Capture VOS properties at multiple granularities
Cascaded human-object interaction recognition (CVPR'20 [1724])
Hierarchical human parsing with typed part-relation (CVPR'20 [1286])
Segmenting objects from relational visual data (TPAMI'21 [810])
Structured scene memory for vision-language navigation (CVPR'21 [1241])
Cross-image pixel contrast for semantic segmentation (ICCV 21 [1285])
Face forensics in the wild (CVPR'21 [1722])
Multi-granularity for instance-aware human semantic parsing (CVPR'21 [1723])
Rethinking Semantic Segmentation: A Prototype View (CVPR [1721])
Deep Hierarchical Semantic Segmentation (CVPR'22 [661])
LIIR (CVPR'22 [662]): Locality-aware inter-and intra-video reconstruction
Visual Abductive Reasoning (CVPR'22 [720])
GMMSeg (NeurIPS'22 [718]): Gaussian Mixture based Generative Seg
Learning Equivariant Seg with Instance-Unique Querying (NeurIPS'22 [1278])
Towards Versatile Embodied Navigation (NeurIPS'22 [1236])
Visual recognition with deep nearest centroids (ICLR23 [1276])
A single agent for instruction following + route creation (CVPR'23 [1294])
ORAN (ICCV'23 [110]): Cross-task navigation skill transfer
LOAFisheye (ICCV'23 [1492]): Fisheye distortions by rotation-equivariant
Cluster3Dseg (ICCV'23 [283]): Clustering for point cloud segmentation
Visual tasks with varying degrees of clustering granularity (NeurIPS'23 [721])

LOGICSEG (ICCV'23 [659]): Semantic neural induction and logic reasoning
DREAMWALKER (ICCV'23 [1237]): Synthesize, assess, interpret plans
LogicDiag (ICCV'23 [719]): Pseudo label conflict via symbolic knowledge
BEV (ICCV'23 [771]): Encode scene layouts and geometric cues via BEV
Prototype-based semantic segmentation (TPAMI'24 [1720])
Clustering for Protein Representation Learning (arXiv'24 [995])
An Omnifit Model for Image Reconstruction from Human Brain Activity
(CVPR'24 [996])
Vision-Language Navigation with Energy-Based Policy (NeurIPS'24 [770])
A Survey on 3D Gaussian Splatting (arXiv'24 [99])
A Survey of World Models for Autonomous Driving (arXiv'25 [284])

10.18 Tianlong Chen

Link.

Lading (ICML'23 [495]): Train multi GNNs combine via greedy interpolation
L2PG (ICML'23 [135]): Predict the fast direction from history
GNT-MOVE (ICCV'23 [179]): A shared permanent expert and a spatial
consistency
Sparse mixture-of-attention and the sparse mixture-of-experts (arXiv'23 [960])
Adaptive MoE framework for multi-task vision recognition (ICCV'23 [123])
Sparse moe with language guided routing for multilingual machine translation
(ICLR'24 [1689])

10.19 Liang Zheng

Link.

Scalable Person Re-Identification: A Benchmark (ICCV'15 [1698])
A video benchmark for large-scale person re-identification (ECCV'16 [1697])
CLAN (CVPR'19 [827]): Category-level feature align under domain shift
Circle Loss (CVPR'19 [1127]): Similarity scores learnt with different paces
Random Erasing (AAAI'20 [1709]): Randomly erases a rectangle region
Rotation prediction has strong linear relation with accuracy (ICML'21 [208])
Distribution difference between train and test reflects acc (CVPR'21 [211])
EI (NeurIPS'22 [209]): Generalize and invariance show strong linear relation
BoP (CVPR'23 [1186]): Train suitability and test difficulty via prototypes

10.20 Bo An, Lei Feng, and Hongxin Wei

Link.

Hard improve calibrated result via post-hoc calibration (NeurIPS'21 [1221])
Taylor CE (IJCAI'21 [282]): Control the Taylor CE to weight samples
Sconf (ICML'21 [75]): Learn unlabeled pairs with similarity confidence
LogitNorm (ICML'22 [1326]): Enforce a constant vector norm on the logits

Open Sampling (ICML'22 [1325]): Introduce OOD instances to re-balance
Inter-class dispersion correlates accuracy and intra-class compactness does
not reflect the generalization performance on OOD (NeuIPS'23 [1416])

10.21 Gao Huang

Link.

Enforce channel-level sparsity in the network (CVPR'17 [796])
Direct connections between any two layers with same size (TPAMI'19 [453])
Semantic data augmentation (NeurIPS'19 [1306] TPAMI'21 [1298])
DvT (NeurIPS'21 [1299]): Not all images are worth 16x16 words
Dynamic Neural Networks: A Survey (TPAMI'21 [386])
Dynamic transformers for efficient image recognition (NeurIPS'21 [1300])
Integration of Self-Attention and Convolution (CVPR'22 [929])

Chapter 11

Experiments Setup

11.1 Dataset

Dataset for different experiments

Digits: MNIST, USPS, SVHN and SYN

USPS (TAPMI'94 [480])

MNIST (IEEE'98 [611])

Caltech-256 (arXiv'07 [338])

Office31 (ECCV'10 [1028]): Amazon, DSLR, and Webcam

ImageNet (CVPR'09 [206], NeurIPS'12 [596], IJCV'15 [1027])

SVHN (NeurIPS'11 [904])

Office Caltech (CVPR'12 [327]): Amazon, DSLR, Webcam, and Caltech

COCO (ECCV'14 [748])

CREAM-D (TAC'14 [68])

Market1501 (ICCV'15 [1699])

Fashion-MNIST (arXiv'17 [1401])

Office-Home (CVPR'17 [1205]): Art, Clipart, Product, and Real World

PACS (ICCV'17 [640]): Photo, Art Painting, Cartoon, and Sketch

VLCS (ICCV'17 [269]): Caltech, Labelme, Pascal, and Sun

CMU-MOSEI (ACL'18 [1574])

Conceptual captions: A cleaned, hypernymed, image alt-text dataset (ACL'18 [1054])

Flickr30k ([1551])

GQA (CVPR'19 [479])

VQAV2 (CVPR'17 [333])

TextVQA ([1094])

Vizwiz ([375])

OCRVQA ([884])

OKVQA ([863])

ScienceQA ([805])

IconQA([806])

AVE (ECCV'18 [1173])
SYN (arXiv'18 [1024])
AVMNIST (ECCVW'18 [1208])
DomainNet (ICCV'19 [962]):
WePerson (ACM MM'21 [648])
SSW60 (SSW'22 [1197])

11.2 Network

Networks for different experiments

PLS (MA'66 [1345]):
Ridge (Tech'70 [419]):
MLP (MCSS'89 [189]): Multilayer Perceptron
LineReg (JW'03 [1041])
Elastic Net ([1773])
LSTM (NC'10 [871])
GRU (arXiv'14 [161])
Lasso(RSSS'96 [1176]): Regression shrinkage and selection via the lasso
OLS (MDOR06 [222]): Ordinary least squares
GoogLeNet (CVPR'15 [1133])
ResNet (CVPR'16 [409])
ResNeXt (CVPR'17 [1417])
MobileNet (arXiv'17 [434])
DenseNet (TPAMI'19 [453])
EfficientNet (ICML'19 [1140])
Fbnet (CVPR'19 [1350]): Hardware-aware efficient convnet design
Convnet (CVPR'22 [798])

11.3 Cluster

Some interesting clustering methods

Nearest neighbors algorithm: (TIT'67 [181])
DBSCAN (SIGKDD'96 [255]): Density-based spatial clustering with noise
Finch (CVPR'19 [1033]): Parameter-free Cluster via first neighbor
Kmeans: (BSMSP'67 [853] SIAM'07 [23])
HAC: (JASA'63 [1320])
CKA: (ICML'19 [594]): Revisit neural network representations similarity
t-SNE (JMLR'08 [1196]): Visualizing data using t-SNE
h-NNE (CVPR'22 [1034]): Hierarchical nearest neighbor graph for dimensionality reduction

11.4 Optimizer

Optimizer

SGD (AoMS'51 [1015])

Adam (arXiv'14 [582])

11.5 Loss

Loss function

Entropy (BSTJ'48 [1051])

KL Loss (AMS'51 [600] and AP'75 [182])

Empirical and structural risk minimization (NeurIPS'91 [1199])

CE Loss (AOR'05 [202])

Triplet Loss (CVPR'15 [1040]):

Center (ECCV'16 [1343]): Penalize distances between feats and class centers

L-Softmax (ICML'16 [779]): Control the adjustable margin

Normface (ACMMM'17 [1226]): Cosine similarity with a scale parameter

Sphereface (CVPR'17 [778]): Softmax loss with angular margin

AT_k (NeurIPS'17 [264]) Average over the k largest individual losses

High confidence on true + uniform distribution on false (NeurIPS'18 [935])

Symm cross entropy for robust learning with noisy labels (ICCV'19 [1305])

MS (CVPR'19 [1290]): Self-similarity and relative similarities of a pair

ArcFace (CVPR'19 [207]): Additive angular margin penalty

Proxy-Anchor (CVPR'20 [575]): Consider relative hardness within batch

Trade-off between invariance and relevance for loss select (NeurIPS'21 [593])

Tilted empirical risk minimization (ICLR'21 [674])

PolyLoss (ICLR'22 [624]): Allocate different polynomial bases importance

Decoupled Kullback-Leibler Divergence Loss (arXiv'23 [184])

11.6 Visualization Analysis

([1196])

Grad-cam: Visual explanations from via gradient-based localizatio (ICCV'17 [1042])

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